Kinetic and Thermodynamic Parameters for the Flux Module

Five predictors run, and formation energies

recomputed with uncertainty

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Abstract

The enzyme-constrained flux layer needs three parameter tables and had almost none of them. Measured k_{cat} covered 148 of 3,728 catalytic units, 4.0 percent, with the other 96 percent running on a single flat default of 10.3 s^{-1}; K_M had no table at all; and the thermodynamics came from a shipped file of point values with no covariance, usable only for this organism. All three are now built, and every value carries the `sha256` of the weights or the model version that produced it.

Six kinetic predictors were mirrored into a standard location with per-file hashes, retrieval commands and git commits, and run over the enzyme-substrate pairs of yeast-GEM 9.0.2, with protein sequences taken from the genome object rather than from the table the model bundles. Validating each against its own authors' published examples paid for itself: **two of these models exponentiate base 2 and three base 10**, and reading either wrong yields turnover numbers that still look credible. They agree on the median k_{cat} and disagree per pair, at a rank correlation of **0.015 to 0.509** on identical pairs. Since almost none of these pairs has a measured value, that spread, and not any published accuracy figure, is the honest uncertainty on a single predicted turnover number.

Formation energies were recomputed from eQuilibrator's component contribution, which returns a covariance and applies to any organism because it works from structure. Summing the resulting table reproduces eQuilibrator's own reaction call to 6.8×10^{-13} kJ mol^{-1}. **The file it is compared against turns out to be in kilocalories and labeled kilojoules**: yeast-GEM's curation script copies yETFL's values across with no conversion, and regressing our table on yETFL's over 1,927 shared metabolites gives a slope of 4.187 at $r = 0.987$. Correcting that and matching the compartment conditions closes the transport arm of the disagreement, median 17.4 to 5.6 kJ mol^{-1}, and leaves the single-compartment cloud where it was, 16.3 to 17.6. Only the second is a disagreement about thermodynamics.

The largest single coverage change in this work was a bug of ours. The proton is the reference of the Legendre transform rather than a compound to be priced, and eQuilibrator correctly declines to give it a formation energy; left as a missing value it removed every reaction that moves a proton. Pricing it at zero, verified against eQuilibrator's own reaction call to 0.0000 kJ mol^{-1}, takes reaction coverage from 42.5 to **66.4 percent** and leaves the union with the shipped table unchanged at 90.0. The translocation term those 170 proton-moving reactions then omit is computed separately and checked against yeast-GEM's own column. Writing structures for the lipids no database indexes prices 347 of the 665 metabolites the recomputation could not reach, taking it to 86.3 percent of reactions and past the shipped column, on one method with one uncertainty.

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Contents

1 Where the parameters were, and where they are now ✗ . . . **3**
1.1 The starting state ✗ . . . 3
1.2 A standard place for a model, with provenance ✗ . . . 3
1.3 Two corrections to the readiness audit ✗ . . . 3
1.4 One environment per predictor ✗ . . . 3
2 The predictors, and what running them taught ✗ . . . **4**
2.1 One module per model, on the node-embedding pattern ✗ . . . 4
2.2 Sequences come from the genome, not from the model's bundled table ✗ . . . 4
2.3 Validation caught a log base ✗ . . . 4
2.4 Embedding on four GPUs ✗ . . . 4
2.5 What a predictor is actually missing is a substrate structure ✗ . . . 4
2.6 What each predictor cost to run ✗ . . . 5
3 Thermodynamics recomputed from eQuilibrator ✗ . . . **5**
3.1 Why the shipped table is a dead end ✗ . . . 5
3.2 What the pipeline actually does, stage by stage ✗ . . . 5
3.3 The largest coverage gap was the proton, and it was our bug ✗ . . . 6

3.4 Where the shipped values came from, and why they disagree ✗ . . . 6
3.5 The uncertainty, which is the point ✗ . . . 9
3.6 Agreement with the shipped table ✗ . . . 9
3.7 Compartment conditions, and what a shift across them means ✗ . . . 9
3.8 Why the table is built per metabolite, and what panel c checks ✗ . . . 11
3.9 The transport term, computed rather than recovered ✗ . . . 11
3.10 Where the method stops ✗ . . . 12
4 Reproducing Wu et al.'s Fig. 3 ✗ . . . 13
4.1 What the original figure claims ✗ . . . 13
4.2 The per-pair predictions behind Fig. 3 were not released ✗ . . . 13
4.3 The log base is not a convention ✗ . . . 13
4.4 The comparison between predictors, which the original does not make ✗ . . . 14
4.5 The comparison per gene, which is the unit the model reads ✗ . . . 14
4.6 Accuracy against measured values, and which database can supply them ✗ . . . 18
4.7 What that correlation is made of ✗ . . . 19
4.8 EZKinetics, the seventh predictor and the only condition-aware one ✗ . . . 21
5 Coverage, route by route ✗ . . . 21
5.1 What the tables cover ✗ . . . 23
5.2 Every value, and the route that produced it ✗ . . . 23
5.3 Mapping a model species onto a molecule ✗ . . . 24
5.4 A pK_a route that does not need the license ✗ . . . 25
5.5 Writing the missing lipid structures ✗ . . . 32
5.6 Composition matching cannot buy coverage ✗ . . . 33
5.7 What is left, and which of it is permanent ✗ . . . 33
5.8 The kinetic tables, tier by tier ✗ . . . 34
5.9 The kinetic ceiling is substrate structure, not the models ✗ . . . 34
5.10 Changelog ✗ . . . 35
6 Where these tables enter the model ✗ . . . 37
6.1 Why these two tables, in two equations ✗ . . . 37
7 What these tables cannot support ✗ . . . 39
7.1 A predicted turnover number is not a measured one ✗ . . . 39
7.2 K_M is built and deliberately not consumed ✗ . . . 40
7.3 The thermodynamic gaps ✗ . . . 40
7.4 What the mirror records when a predictor cannot run ✗ . . . 40
8 Every source, for citation and for the mirror ✗ . . . 40
8.1 Papers, by the parameter they supply ✗ . . . 40
8.2 Code and weights ✗ . . . 41
8.3 Five provenance defects to repair ✗ . . . 41

1 Where the parameters were, and where they are now ✗

1.1 The starting state ✗

Before this work, yeast-GEM 9.0.2 [1] had 3,728 catalytic units and one turnover number for almost none of them. Measured k_{cat} covered 148 units, 4.0 percent, drawn from the Open Enzyme Database's [2] 1,126 *S. cerevisiae* rows, which between them name only 106 distinct UniProt accessions. The other 3,580 units, 96 percent of the model, ran on a single flat default of 10.3 s^{-1}, the median of the values that did resolve. K_M had no table at all: the parameter existed as an enum member and a protocol flag with nothing behind either.

Thermodynamics was in better shape and worse than it looked, which Section 3 takes up.

1.2 A standard place for a model, with provenance ✗

The predictors arrived as ad-hoc git clones in a scratch directory, which is not something a result can be rebuilt from: nothing recorded which commit was cloned, and nothing would detect an upstream force-push. They now live at `$DATA_ROOT/models/kinetics/<name>/`, beside the existing `mineru` mirror, each with a `manifest.json` pinning every file by `sha256` and recording the source URL, the exact retrieval command, and the git commit.

Two trees are kept deliberately apart. `$DATA_ROOT/models/...` holds *the model*: weights, vocabularies, dictionaries, written once and verified thereafter. `$DATA_ROOT/data/torchcell/...` holds *what the model produced*, regenerable from a mirror plus code, and every row carries the `sha256` of the weight file that produced it. Deleting a build is routine; deleting a mirror invalidates the provenance of every build made from it, so a mirror is only ever added.

A value with no provenance becomes indistinguishable from a measured one within two steps, and the flux layer's coverage gate has to tell an experimental value, a predicted value and a filled default apart.

1.3 Two corrections to the readiness audit ✗

An earlier audit of these six repositories recorded which ship weights. Reading the files rather than the filenames reversed two of its entries, and both reversals matter.

DLKcat does ship its trained model. It is `Results/output/all--radius2--...--iteration50`, a PyTorch `state_dict` with no file extension, which is why a scan for `*.pt`/`*.pkl` missed it. On the strength of that miss an 18 GB Zenodo download had been queued that was never needed.

Boost_KM ships two fitted models, and the citation points at the wrong one. Its 18 "weight" files are the UniRep 1900 mLSTM *featurizer*, and reading only those supports the conclusion that no regressor was released. Beside them sit `xgboost_model.dat` and `xgboost_model_full.dat`, which are the fitted K_M regressors, and the repository's own README says the pretrained weights can be loaded rather than retrained. Nothing needed refitting.

What Wu et al. ran is a third model again. Their Methods cite `AlexanderKroll/KM_prediction`, the repository for the published paper, but the notebook in their release vendors `KM_prediction_function`, whose enzyme representation is ESM-1b rather than UniRep and whose regressor, `xgboost_model_new_KM_esm1b.dat`, takes 1,332 features rather than 1,952. Kroll's README states the substitution plainly: "the model provided in the repository 'KM_prediction_function' is slighly different to the one presented in our paper: Instead of the UniRep model, we are using the ESM-1b model." The substrate encoder is common to both, and its three checkpoint files are byte-identical across the two repositories, so only the enzyme half and the regressor differ. Reproducing Fig. 3f means following the code rather than the citation.

1.4 One environment per predictor ✗

A released predictor is a frozen artifact, and several of them pin dependencies that cannot coexist. UniKP's regressors were pickled before scikit-learn 1.3 added `missing_go_to_left` to the decision-tree node dtype, and scikit-learn 1.7 refuses to load them. Refusing is correct: reinterpreting a tree's node array under a changed dtype would yield numbers that are not UniKP's predictions.

So UniKP runs in an environment pinned to scikit-learn 1.2.2, invoked as a subprocess with parquet as the interface, and the torchcell-side dataset class calls it rather than importing it. DeepEnzyme additionally wants NVIDIA apex and an old numpy for the same reason.

2 The predictors, and what running them taught ✗

2.1 One module per model, on the node-embedding pattern ✗

Which predictor feeds the flux layer is a design choice to experiment over, not a pipeline detail to settle once, so the kinetic predictors copy the shape the sequence-embedding datasets already use: an abstract base, one module per model, and a name-to-class registry that a config string selects.

One structural difference drives everything else. A node embedding is keyed by *gene*; a kinetic parameter is keyed by a *pair*, because a turnover number is a property of one enzyme acting on one substrate. The key is (uniprot, substrate, predictor, parameter). Collapsing to one row per gene would silently pick a substrate, and collapsing to one row per catalytic unit would silently pick a subunit, so the store holds pairs and aggregation lives in the resolver.

2.2 Sequences come from the genome, not from the model's bundled table ✗

yeast-GEM ships a `swissprot.tsv` with a sequence column, and it sits conveniently beside the model. It is the wrong source. Protein sequences are read from `SCerevisiaeGenome` through `genome[orf].protein.seq`, the same path the ESM2 embedding dataset uses, so everything stays on the derivation chain from the S288C reference FASTA plus the GAF. The difference is provenance rather than content: a sequence from a third-party table is a claim by that table's compiler, while the same sequence from the genome is the one the perturbation ontology edits, so a strain's genotype and the protein a predictor sees cannot silently disagree.

All 1,161 yeast-GEM genes resolve as `current` through the shared gene-name reconciler, and all 1,161 have a protein sequence, so nothing is lost by the switch. Rebuilding DLKcat on genome sequences moved the median k_{cat} from 4.0125 to 4.0116 s^{-1}.

Two failures are worth recording because both are silent. `orf in genome` looks safer than indexing and is not: the genome defines no `__contains__`, so Python falls back to integer indexing and calls `genome[0]`, raising before the real lookup happens. And `go_root` is a separate constructor argument defaulting to a relative path; left unset it misses and the constructor tries to fetch `go.obo` from `current.geneontology.org`, which answers 403.

2.3 Validation caught a log base ✗

The DLKcat implementation reproduces the authors' three released example values to four decimal places, and that check caught two defects that would otherwise have been invisible in a plausible-looking table. The first is the exponentiation base, which Section 4.3 tabulates for all six predictors.

Multi-component SMILES are refused by the model itself. Its fingerprint pass indexes a node list built only from bonded atoms, so a disconnected fragment, such as the lone `[Fe+2]` in the heme SMILES yeast-GEM gives for ferricytochrome *c*, walks off the end of the list.

2.4 Embedding on four GPUs ✗

UniKP's protein half is ProtT5-XL [3], a 3-billion-parameter encoder, and its pinned environment holds CPU-only torch. The embedding is therefore produced once on GPU and handed across as a content-addressed cache, keyed by a hash of the sequence so that any predictor wanting a ProtT5 representation can reuse it and two ORFs with one sequence resolve to one vector.

Sequences are split by index across the four GPUs and merged; within a shard they are sorted by length and batched, which is where most of the gain comes from, and a per-sequence masked mean keeps the result identical to unbatched inference. The 1,154 distinct sequences finish in under 40 seconds.

The genome is read exactly once, by a dump step, and the shards read that dump. Four processes opening the same gffutils SQLite database concurrently raced and made it briefly unreadable; since the shards need the genome for nothing but a sequence lookup, the dependency was removed rather than serialized.

2.5 What a predictor is actually missing is a substrate structure ✗

A predictor takes a protein sequence and a substrate structure. Every gene has a sequence, so nothing a predictor fails to price fails for want of a protein, and an audit of the 176 catalytic units no admitted predictor could price found the

same blocker on all 176: no structure for any substrate the reaction consumes. None was blocked by an accession, a sequence, the complex rule or a predictor crash.

Three routes recovered 165 of the 176, and each of them is a mapping rather than a model. An annotation cross-reference priced 27, following the identifier yeast-GEM carries through MetaNetX's own cross-reference table to a current entry: the model annotates an older MNXref release, so `MNXM417` for fructose 1,6-bisphosphate is not a row in MNXref 4.5 and three reactions had no turnover number for that reason alone. The lipid structure writer of Section 5.5 supplied 48 more from structures it had already composed for the thermodynamics. A substrate writer built the remaining two by constructing every chain length and keeping the one whose RDKit formula equals the model's, which selects dolichol-4 uniquely for dolichyl D-mannosyl phosphate out of $n = 2$ to 11. Eleven units none of the three reached.

Acceptance is composition, not similarity. A candidate structure is admitted only when its RDKit formula equals the model's formula with both moved to the neutral species by adding one hydrogen per negative charge. That rejects an R-group placeholder outright, since `C4H7O2SR` never equals `C4H7O2*`, and it is what caught the structures the tables were already resting on: of the 1,425 distinct substrate structures the first round of predictions used, **97 fail the check**, 35 of them tRNA species the model gives the formula `RH`. Repairing 31 and dropping the rest cost 8 catalytic units and moved none of the surviving values by more than one decade for k_{cat}.

Two table versions are kept and they are not interchangeable. The `processed` tables are the first round, 3,717 units on unchecked structures; the `verified` tables are 3,709 units, 1,359 distinct structures and none failing the composition check. Every predictor was rerun on the verified pairs and both versions are hashed per predictor and per parameter in `results/kinetic_table_versions.json`, so a stated tier count names the version it belongs to. Section 5.8 reports the verified ones.

2.6 What each predictor cost to run ✗

Table 1: The six predictors named in Wu et al.'s Methods, what each emits, and what it took to run here. "Ships weights" refers to the repository as cloned; UniKP's fitted regressors are released separately on HuggingFace and are mirrored alongside it.

Predictor	Emits	Ships weights	Note
DLKcat	k_{cat}	yes, extensionless	output is $\log_2$
UniKP	k_{cat}, K_M	regressors on HuggingFace	needs scikit-learn 1.2.2
TurNuP	k_{cat}	Zenodo archive	takes full reaction SMILES
EITLEM-Kinetics	k_{cat}, K_M	Zenodo, 12.3 GB	many checkpoints
Boost_KM	K_M	yes, two regressors	ESM-1b variant is the one Wu ran
DeepEnzyme	k_{cat}	figshare	uses AlphaFold structures

3 Thermodynamics recomputed from eQuilibrator ✗

3.1 Why the shipped table is a dead end ✗

yeast-GEM 9.0.2 ships `model_metDeltaG.csv` and `model_rxnDeltaG.csv`, and they are usable: 2,389 of 2,806 metabolites and 3,210 of 4,131 reactions carry a value. They also close two doors. They are point values with no covariance, so the correlated error term $\Delta_r G^{\circ,\text{error}} = Qm$ that a thermodynamic flux formulation calls for cannot be formed from them at all. And they exist only for this model, so a table built on them transfers to no other organism.

Component contribution answers both at once: it decomposes a compound into groups, fits group energies against the measured reaction corpus, and returns a formation energy with the covariance implied by that fit, from a structure alone.

The shipped files also use *two* missing-value conventions: most gaps are the sentinel 10^7, but a further 51 metabolites and 120 reactions are the literal string `NaN`. Filtering only the sentinel leaves the NaNs to propagate through every sum, producing NaN gradients from what looks like a coverage of 87 percent.

3.2 What the pipeline actually does, stage by stage ✗

Each metabolite is resolved by accession, MetaNetX first, then ChEBI, then KEGG, then BiGG, with an InChI structure lookup as a fallback (Figure 1a). A metabolite becomes a formation energy in four stages, each of which loses some. An identifier has to resolve to an entry in eQuilibrator's compound cache; that entry has to carry a group decomposition; the fit returns an untransformed $\Delta_f G^{\circ}$; and a Legendre transform to (pH, ionic strength, temperature, pMg) turns it into

the $\Delta_f G'^{\circ}$ the layer consumes. Reaction energies are then $\sum_i S_{ij}\Delta_f G'^{\circ}_i$, so a reaction survives only if *every* participant does, which is what makes reaction coverage the binding number and a metabolite-level figure an overstatement.

That is the first of eleven stages, and every one of them writes a route or a reason. `thermo_pipeline_yeast9.py` runs curated records, the eQuilibrator recomputation, an accession sweep, the external pK_a targets, the lipid structure writer, the ion formation table, the composition surrogate, the polymer repeat-unit surrogate and the shipped metabolite column in that order, then two diagnosis stages whose only job is to say why the remainder is unpriced. A later stage may overwrite an earlier stage's *reason* but never its value, with one exception: a reason a curator recorded after running the group decomposer is final, so a measured decomposition failure cannot be overwritten by a namespace heuristic reporting the same species as merely lacking a structure. Section 5.2 reads the resulting attribution out route by route, and the rest of this section is the recomputation stage alone.

3.3 The largest coverage gap was the proton, and it was our bug ✗

The recomputation covered 1,755 reactions, 42.5 percent, against the shipped table's 84.9, and that shortfall was read as the price of an organism-independent method. Ranking metabolites by how many otherwise-priceable reactions each blocks says otherwise: the top twelve were all $\mathbf{H}^+$, in twelve compartments, together blocking 1,477 reaction slots.

`standard_dg_formation` returns nothing for the proton, and that is correct rather than a failure. A bare proton has no group decomposition, and its chemical potential is what the Legendre transform to a fixed pH is defined *against*: the proton is the reference of the transform, not a compound to be priced by it.

The convention is that the transformed proton contributes zero, and it was verified rather than assumed. On four proton-carrying, mass-balanced reactions, alcohol, lactate and malate dehydrogenase and glucose-6-phosphate dehydrogenase, the sum $\sum_i S_{ij}\Delta_f G'^{\circ}_i$ with the proton term set to zero reproduces eQuilibrator's own `standard_dg_prime` to $\mathbf{0.0000\,kJ\,mol^{-1}}$. The whole-table check still passes afterwards: over 40 single-compartment balanced reactions the two routes agree to $6.8 \times 10^{-13}\,\mathrm{kJ\,mol^{-1}}$.

Reaction coverage goes from 1,755 to 2,741, a gain of 986 reactions and 23.9 percentage points, from one line. The union with the shipped table does not move at all, staying at 3,716: every one of those 986 reactions was already covered by the GEM. What changed is how much of the coverage two independent methods confirm, which is what makes a disagreement between them detectable.

One caveat, and it is a wrong number rather than a missing one. Fixing the proton at zero is exact within a single compartment, which is what the validation tested, and not for a proton moved *between* compartments at different pH, where the translocation term is now zero by construction. **183 reactions move a proton across a membrane and 170 of them are now priced**, each missing that term. Before the fix they were unpriced; now they carry a value that is confidently wrong. That term is supplied separately, and Section 3.9 is where it is computed and checked.

The two tables are not nested, and that changes what to do with them.

Table 2: Which of yeast-GEM's 4,131 reactions each table can price, comparing the recomputation against the GEM's metabolite column summed over participants, which is the like-for-like comparison since both are then metabolite tables.

Priced by	Reactions	Share
both	2,538	61.4%
eQuilibrator only	**203**	**4.9%**
shipped only	967	23.4%
neither	423	10.2%
union of the two reaction sets	3,708	89.8%
priceable under the merged metabolite table	3,716	90.0%

Those last two rows differ by 8 reactions, and the difference is real. Merging the two metabolite tables first is larger, because a reaction can draw one participant from each while neither route prices it alone.

The 203 reactions only the recomputation prices are led by thioredoxin, palmitoyl-CoA, FMN and riboflavin, chemicals the shipped table simply lacks. The 967 it loses are the real cost. They are not concentrated (Figure 1e), and the blockers are the same phosphatidylcholine and phosphatidylethanolamine species that block the thermodynamics everywhere else. Writing structures for that population is what Section 5.5 does.

3.4 Where the shipped values came from, and why they disagree ✗

Yeast9 almost certainly used a group-contribution method too, and the disagreement is still expected, for three reasons read out of the upstream artifacts rather than inferred from the values. **The shipped column is in kcal per mole and is labeled kilojoules; its metabolite half was flattened across compartments; and its estimator is**

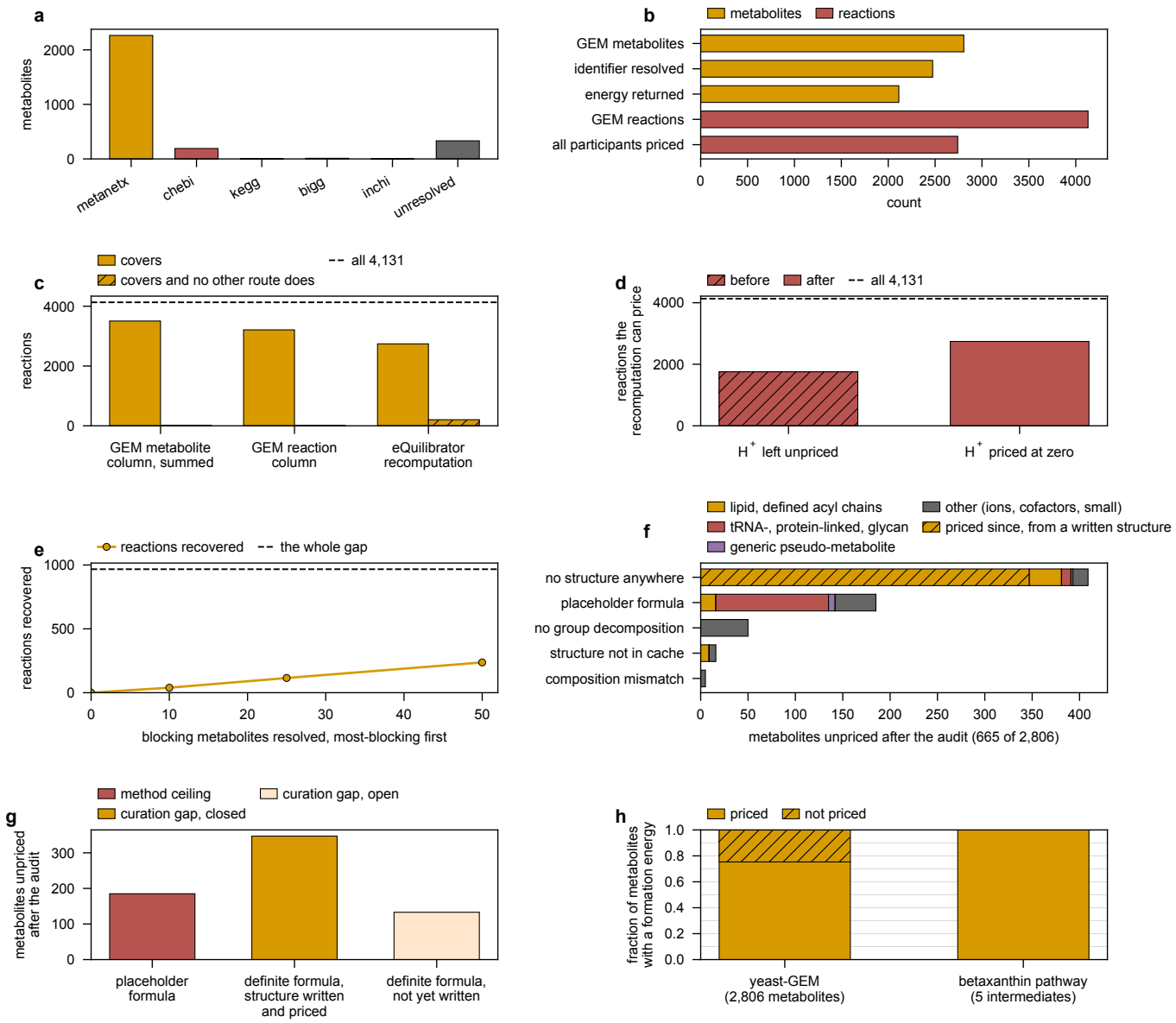


Figure 1: What the recomputation does to yeast-GEM, what it covers, and where the remaining coverage is. a, Which identifier namespace resolved each of the 2,806 metabolites to an eQuilibrator compound, in the order the resolver tries them, so the bars are disjoint: a metabolite under `chebi` is one whose MetaNetX identifier missed. MetaNetX 2,265, ChEBI 191, BiGG 11, KEGG 5, InChI fallback 2, and 332 unresolved. **b**, The pipeline as a funnel, on the built table the layer reads. 2,474 metabolites resolve an identifier to the compound cache, 2,113 of those carry a group decomposition and get an energy, and 2,741 of the 4,131 reactions have every participant priced. Three recovery routes in the audit, extra namespaces, MetaNetX cross-references and an InChIKey structure lookup, add 28 metabolites and 46 reactions that are not yet built into the table. **c**, Three routes to a reaction energy, what each covers and, hatched, what only it covers. yeast-GEM ships *two* independent columns built from different sources; the recomputation is a third and is the only route for 203 reactions. **d**, The proton convention, worth 986 reactions on its own: 1,755 before it, 2,741 after. **e**, Whether a short list of metabolites would recover the rest. Resolving the 50 most-blocking metabolites recovers 237 of the 967 reactions the shipped table prices and the recomputation cannot, so the gap is diffuse. **f**, What still blocks each of the 665 metabolites the audit left unpriced, and what kind of metabolite it is. Every panel is the recomputation stage alone; the built table is Figure 11. The largest bar is one population, lipid species carrying a definite acyl composition with no identifier in any namespace the model carries. **Hatched, the 347 of those 409 the lipid structure writer has since priced** (Section 5.5); those values are not in the table panels a to e are computed on, being all at pH 7.2. The placeholder bar is the opposite case, entries whose formula contains an R, X or $*$: a GEM writes these because the reaction is real and the substrate is a class rather than a molecule, so no structure exists to protonate. **g**, The same 665 split into what the method cannot do and what curation has not done. A placeholder formula is a ceiling, 185 entries. The 480 with a definite formula are a curation gap, and 347 are now closed, leaving 133 open. **h**, The heterologous test case. 75.3 percent of yeast-GEM's own metabolites carry a formation energy against 5 of the 5 betaxanthin pathway intermediates, all resolved from the compound cache by InChIKey prefix rather than by the exact-InChI match that read 2 of 5 (Section 5.4). Generated by `experiments/026-metabolism-flux/scripts/plot_thermo_pipeline.py` from `audit_reaction_thermo_coverage.py`, `audit_unresolved_metabolites.py`, `lipid_structure_writer.py` and `pathway_thermo.py`; every count is written back to `results/thermo_pipeline_figure.json`.

a different group-contribution method fitted to a different corpus. The first two are bookkeeping errors in yeast-GEM. The third is the genuine method difference and is the smaller effect.

Two names recur below. **yETFL** is the expression-and-thermodynamics model of yeast (Oftadeh et al. 2021), which supplies almost every shipped value. **matTFA** is the MATLAB thermodynamic-flux toolbox that produced yETFL's thermodynamic fields, and its conventions are what the shipped columns inherit.

The chain, read out of the curation script ✗

yeast-GEM's own `code/modelCuration/v8_7_0.m` (PR #330) assigns both columns.

Table 3: How yeast-GEM's two thermodynamic columns were populated. Three methods sit behind one model, and the metabolite column and the reaction column were built independently of each other. yETFL supplies 3,177 of the 3,210 reaction values; dG-predictor supplies the remaining 33.

Column	Source, in preference order	Matching key
`metDeltaG`	yETFL, then ModelSEED	metabolite *name* with the compartment tag *stripped*; then `kegg.compound` id
`rxnDeltaG`	yETFL, then dG-predictor	reaction id; then **row position**, assigning `dgpred(3983:4062)` to `rxns(3983:4062)`

yETFL states its own method, Mavrovouniotis (1990) as packaged in matTFA's `thermo_data.thermodb` with Alberty's transform (Table 27); the fallback, dG-predictor (Wang et al. 2021), is a Bayesian ridge over radius-1 and radius-2 molecular signatures at pH 7.0, ionic strength 0.1 M and 298.15 K, reporting kilojoules.

The shipped column is kcal per mole ✗

yETFL's model carries the field `thermo_units = 'kcal/mol'`, and `v8_7_0.m` copies its values across with a `round(x,2)` and no conversion. The yeast9 paper reports the resulting columns in $\mathrm{kJ\,mol^{-1}}$.

This is measured, not read off a label. Regressing our transformed formation energies, which are in $\mathrm{kJ\,mol^{-1}}$, on yETFL's over the 1,927 metabolites both price gives a slope through the origin of **4.187** at Pearson $r = 0.987$, against the thermochemical $4.184\,\mathrm{kJ\,kcal^{-1}}$. Individual compounds settle it beyond argument: yETFL prices cytosolic water at −37.21 and ATP at −564.41 against Alberty's tabulated −157.6 and $-2292\,\mathrm{kJ\,mol^{-1}}$. Read as kcal those are −155.7 and −2361, within 1 and 3 percent; read as kJ they would be wrong by a factor of four. The $69\,\mathrm{kJ\,mol^{-1}}$ still separating yETFL's ATP from Alberty's is a single-compound estimation difference, and it is the size of difference the single-compartment cloud in Figure 3b is made of. The reaction column inherits the unit, being yETFL's reaction field verbatim: on the 3,177 reactions in both, the largest difference is 5.0×10^{-3}, which is exactly what `round(x,2)` leaves. The 33 reactions that are *not* yETFL's are the dG-predictor block, which is in kilojoules, so the shipped reaction column mixes two unit systems.

The metabolite column was flattened across compartments ✗

matTFA transforms at each compartment's own pH and ionic strength, so yETFL holds a different value per compartment, while `v8_7_0.m` matches on the compartment-stripped name, so every copy receives whichever compartment survived the de-duplication.

Half of the shipped metabolite column is therefore the value for a different compartment: **1,162 of 2,324** entries match their own compartment's yETFL value and the rest do not, with a 75th-percentile error of **86.7** and a mean of **74.5** $\mathrm{kJ\,mol^{-1}}$. Those are large because the transform's pH dependence scales with a compound's proton count: ATP carries about twelve protons at pH 7, so moving it two pH units moves its transformed formation energy by more than $130\,\mathrm{kJ\,mol^{-1}}$.

This accounts for the residual the flux layer records ✗

The flux layer's median disagreement of 10.2 between the summed metabolite column and the shipped reaction column is yeast-GEM disagreeing with itself, not with eQuilibrator.

Within yETFL's own model the identity is exact: over the 2,677 single-compartment reactions it prices, $\sum_i S_{ij} \Delta_f G'^{\circ}_i$ with the proton dropped reproduces its reaction field to a maximum absolute residual of **0.0**.

Table 4: Median absolute residual between $\sum_i S_{ij}\Delta_f G_i'^{\circ}$ and yeast-GEM's shipped reaction column, in kcal mol^{-1}, which is the unit both columns are in. Two changes remove it: drop the proton from the sum, and use yETFL's per-compartment values in place of the flattened column. matTFA prices the proton at $RT\ln(10)\,\mathrm{pH} = 9.54$ kcal mol^{-1} and then never adds it into a reaction sum, because the transform already carries the reaction's pH dependence; yeast-GEM ships that number in the metabolite column anyway, so a consumer that sums the column pays it once per net proton.

Metabolite values summed	Proton	Single compartment	Transport
shipped column	included	10.21 (n=2,021)	2.91 (n=1,183)
shipped column	dropped	2.37	2.78
yETFL per compartment	included	9.54 (n=2,005)	0.0039 (n=1,169)
yETFL per compartment	**dropped**	**0.0024**	**0.0038**

3.5 The uncertainty, which is the point ✗

Of 1,673 estimable reactions, 888 have a standard error of exactly zero and the remaining 785 carry a median of 2.30 kJ mol^{-1} and a 95th percentile of 6.01 (Figure 2).

Estimability is a separate question from uncertainty and is easy to conflate. Component contribution returns directions with finite variance and directions with infinite variance, the second being what the training corpus never determined. Almost every compound has a nonzero infinite component, since an absolute formation energy is defined only up to element conservation, and those directions cancel in a mass-balanced reaction, so the test applies to a reaction and never to a compound in isolation. A reaction whose infinite component does not cancel is masked rather than given a finite error bar.

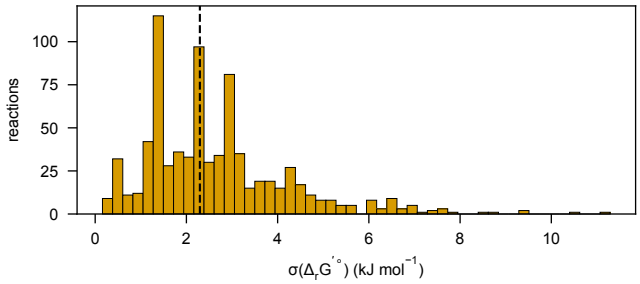


Figure 2: Standard error on $\Delta_r G'^{\circ}$ for the 785 estimable reactions whose uncertainty is nonzero. The dashed line is the median, **2.30 kJ mol^{-1}**. The 888 estimable reactions omitted here have a standard error of exactly zero because they are transport steps whose per-compound uncertainties cancel. The shipped table carries no uncertainty of any kind, so this quantity has no counterpart in it. Generated by `experiments/026-metabolism-flux/scripts/plot_equilibrator_thermo.py`.

3.6 Agreement with the shipped table ✗

Drawn as published, the disagreement is a cross (Figure 3a). Correcting the unit and rebuilding our table at yETFL's own conditions separates the two arms.

The transport arm closes. On the 752 transport reactions the shipped column prices above 0.1, the median absolute difference falls from 17.4 to **5.6** kJ mol^{-1} and Pearson r rises from 0.30 to **0.75**, and over all 1,023 it is 0.57. That arm was two builds holding different compartment conditions rather than a disagreement about thermodynamics.

The single-compartment cloud does not close, and it should not. Its median absolute difference moves from 16.25 to **17.60** kJ mol^{-1} and Pearson r from 0.537 to 0.513. That residual is the genuine estimator difference, and its size is unsurprising: yETFL's value for glucose-6-phosphate is about 39 kJ mol^{-1} from Alberty's, and errors of that size on individual compounds cancel within a reaction only when the compounds share groups. Only the recomputation carries an uncertainty that shows this without the archaeology.

3.7 Compartment conditions, and what a shift across them means ✗

A compartment enters the thermodynamics through three numbers: its pH, its ionic strength, and its electrical potential relative to the cytosol (Figure 4a).

No number in that table traces to a measurement in the library, and the literature column that should sit beside it cannot be written. The pipeline's pH values are textbook organellar figures, recorded in the build output as `assumed`. The only in-library sources for any of them are two modeling papers that tabulate them as parameters rather than measure them: Thermo-Flux's Table 1, which covers the cytosol, the mitochondrion and the extracellular

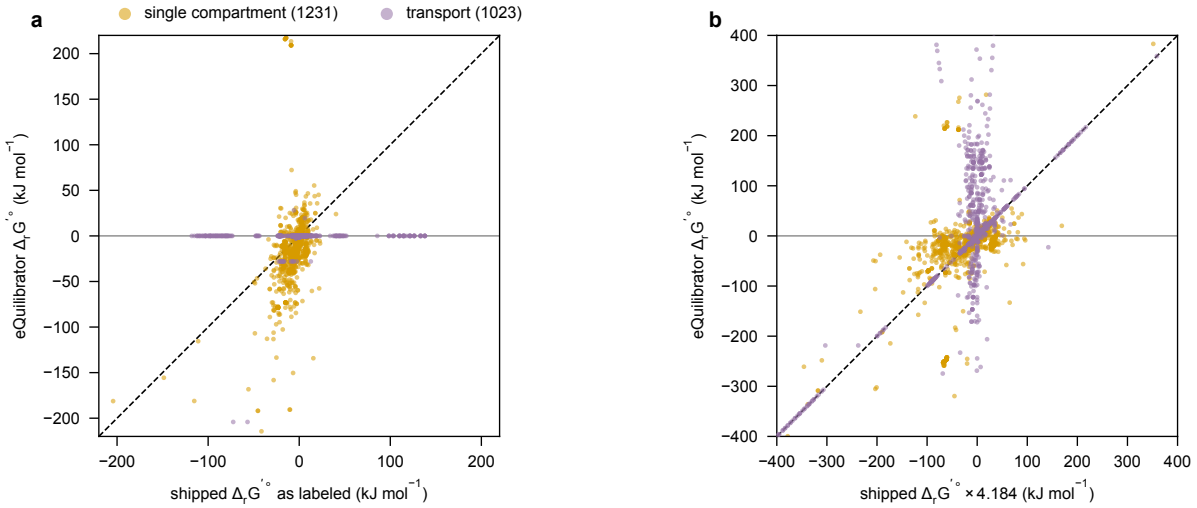


Figure 3: The disagreement with yeast-GEM's shipped reaction column, before and after two bookkeeping corrections. Both panels show the reactions both tables price, 2,254 in **a** and 2,253 in **b**, colored by whether they cross a compartment boundary; the dashed line is equality. **a**, As published: the shipped values read in the kJ mol^{-1} the yeast9 paper states, our table at a uniform pH 7.0, ionic strength 0.25 M and 303.15 K. The horizontal arm at zero is the 1,023 transport reactions, whose formation energies cancel term by term at one pH. On the 1,231 single-compartment reactions the median absolute difference is 16.25 kJ mol^{-1} at Pearson $r = 0.537$. **b**, Corrected: the shipped values multiplied by 4.184, since that column is yETFL's and carries `thermo_units = 'kcal/mol'`, and our table rebuilt at yETFL's own conditions, 298.15 K with each compartment's pH and ionic strength and no magnesium. The transport arm collapses onto the diagonal, median 0.57 kJ mol^{-1} over all 1,023 and 5.6 over the 752 the shipped column prices above 0.1, at $r = 0.75$; the single-compartment cloud does not move, median 17.60 at $r = 0.513$. Note the wider axis range in **b**: correcting the unit widens the shipped axis fourfold. Generated by `experiments/026-metabolism-flux/scripts/plot_equilibrator_thermo.py` from `yeast9_delta_g_provenance.py -recompute`.

Table 5: The conditions each build applies, beside those yeast-GEM 9's shipped values rest on. Potentials are signed against the cytosol, so a negative entry is a compartment held below the cytosol: the mitochondrial matrix, and in yETFL also the nucleus. Thermo-Flux's Table 1 gives magnitudes rather than signs, so the signs in the pipeline column are assigned here from the direction of the gradients they describe. The pipeline runs every compartment at ionic strength 0.25 M, yETFL runs four at 0.25 and the rest at zero. yeast-GEM 9 has no intermembrane-space compartment; `im` appears in yETFL only.

		this pipeline			yETFL, behind yeast9		
	Compartment	pH	I (M)	Φ (mV)	pH	I (M)	Φ (mV)
`c`	cytoplasm	7.2	0.25	0	7.00	0.25	0
`ce`	cell envelope	5.0	0.25	+60	7.00	0	0
`e`	extracellular	5.0	0.25	+60	5.00	0	+60
`er`	endoplasmic reticulum	7.1	0.25	0	7.20	0	0
`erm`	endoplasmic reticulum membrane	7.1	0.25	0	7.20	0	0
`g`	Golgi	6.6	0.25	0	6.35	0	0
`gm`	Golgi membrane	6.6	0.25	0	6.35	0	0
`im`	mitochondrial intermembrane	n/a	n/a	n/a	7.00	0	0
`lp`	lipid particle	7.2	0.25	0	7.00	0.25	0
`m`	mitochondrion	7.8	0.25	−160	7.50	0.25	−180
`mm`	mitochondrial membrane	7.8	0.25	−160	7.50	0.25	−180
`n`	nucleus	7.2	0.25	0	7.00	0	−15
`p`	peroxisome	7.0	0.25	0	8.20	0	0
`v`	vacuole	6.2	0.25	0	6.17	0	0
`vm`	vacuolar membrane	6.2	0.25	0	6.17	0	0

space and nothing else, and yETFL through its compartment-data file. The measurements that would replace them are a curation request: an in vivo cytosolic and mitochondrial pH for *S. cerevisiae* by ratiometric pHluorin (Orij et al. 2009), a vacuolar lumen pH (Brett et al. 2005; Plant et al. 1999), organelle pH by targeted fluorescent protein for the Golgi and endoplasmic reticulum (Llopis et al. 1998), and a peroxisomal measurement, where the two tables in use differ by 1.2 pH units and nothing in the library settles which is right.

Applying a per-compartment pH rather than a uniform one moves 2,220 of the 2,741 covered reactions (Figure 4b). **A shift of that size is what the Legendre transform must produce, and it is not evidence against the method**, by the proton-count dependence that flattens the shipped column above.

3.8 Why the table is built per metabolite, and what panel c checks ✗

The table stores per-metabolite formation energies and forms reaction energies as $\sum_i S_{ij}\Delta_f G'^{\circ}_i$, rather than asking eQuilibrator for each reaction directly. The reason is the flux layer, and it is a real one: in its anchored mode the reaction potential is $\Delta_r G_j = \Delta_r G'^{\circ\prime}_j + RT\sum_i S_{ij}\ln c_i + \Delta_r G^t_j + (Qm)_j$, with $\ln c$ a learned per-metabolite vector squashed into the physiological window, and in its free mode it is $\sum_i S_{ij}\mu_i$ with μ unconstrained. Both need a quantity indexed by metabolite. A reaction-level $\Delta_r G'^{\circ\prime}$ cannot be decomposed back into one, and the square-root covariance factor Σ that gives Q is per metabolite as well.

Nothing about the transform itself is reimplemented. The transform is `Compound.transform`, eQuilibrator's own, and the formation energy is `ComponentContribution.standard_dg_formation`. What is ours is the assembly, and each part of it can be wrong in a way that produces a reasonable-looking table, so panel c checks it against eQuilibrator's own reaction-level call (Figure 4c).

As an absolute configuration check, four textbook reactions price at pH 7.2, ionic strength 0.25 M and 303.15 K as: ATP hydrolysis -28.93 ± 0.30, phosphoglucose isomerase $+2.66 \pm 0.38$, fumarase -3.43 ± 0.28 and triosephosphate isomerase $+5.61 \pm 0.55$ kJ mol^{-1}.

3.9 The transport term, computed rather than recovered ✗

Because the same chemical appears on both sides, $\sum_i S_{ij}\Delta_f G'^{\circ}_i = 0$ over a transport reaction priced at one pH, so its entire standard energy is its transport term and a metabolite-column sum asserts that moving a proton up a two-unit pH gradient is free. Thermo-Flux gives that term as

$$\Delta_r G^t_j = -N_H\, RT\ln(10)\,\Delta\mathrm{pH} - Fq\,\Delta\Phi, \tag{1}$$

with N_H the protons moved across the membrane and q the charge moved. `torchcell/metabolism/transport.py` evaluates it per reaction from the stoichiometry, the compartment of each metabolite and its charge.

The paper does not fix which side each difference is taken from, so the module fixes it and derives the signs. A species moves out of the compartment it appears in as a reactant and in to the one it appears in as a product; the proton potential $\mu^\circ - RT\ln(10)\,\mathrm{pH}_X$ and the electrical work $+qF(\Phi_{\mathrm{in}} - \Phi_{\mathrm{out}})$ then give $\Delta\mathrm{pH} = \mathrm{pH}_{\mathrm{in}} - \mathrm{pH}_{\mathrm{out}}$ and $\Delta\Phi = \Phi_{\mathrm{out}} - \Phi_{\mathrm{in}}$. This is the sign matTFA implements, and choosing it the other way would invert every proton pump in the model without producing any other symptom.

Identifying which species move is the part that needs care. A chemical is matched by name, because yeast-GEM gives the same chemical a different `s_XXXX` identifier in every compartment, and copies in the same compartment on both sides are canceled before the remainder is paired across compartments. Mitochondrial ATP synthase is the worked case: yeast-GEM writes it as three cytosolic protons consumed against two matrix protons produced, so one proton is chemistry and $N_H = 2$ translocate.

N_H counts the explicit H^+ species yeast-GEM carries per compartment and nothing else. Thermo-Flux's N_H also absorbs the protons released or bound when a reactant's average protonation differs between two compartments at different pH, which needs a species distribution for every transported compound. That part is not computed here and is the next term to add.

Against yeast-GEM's own curated column the computed terms track but do not reproduce it. Of 1,468 reactions crossing a compartment boundary, 326 receive a nonzero term, 183 move a proton and 482 move charge. On the 287 reactions carrying a nonzero value in both, with the shipped column converted to kJ mol^{-1}, the sign agrees on 84.7 percent, Pearson $r = 0.804$, and the median absolute difference is 7.04 kJ mol^{-1} against a median magnitude of 12.0. Rebuilding under yETFL's own compartment conditions rather than the pipeline's tightens it slightly, to $r = 0.813$ and a median difference of 6.08 over 299 reactions, which says the residual is mostly method rather than conditions.

Where they disagree is informative. 613 reactions carry a nonzero shipped value and a zero computed one, which is the protonation term left out above. Eleven go the other way, computed nonzero against a shipped zero, and those are reactions yETFL declined to price at all rather than priced as zero.

`transport_delta_g_source` takes `off`, the default and what every arm here ran; `shipped`, which recovers the term from the GEM's reaction column where the metabolite-sum energy is structurally zero, converted out of kcal at parse time by `_read_delta_g_csv`; and `computed`, which reads this module's table and is the only route that carries to a model shipping no thermodynamics of its own.

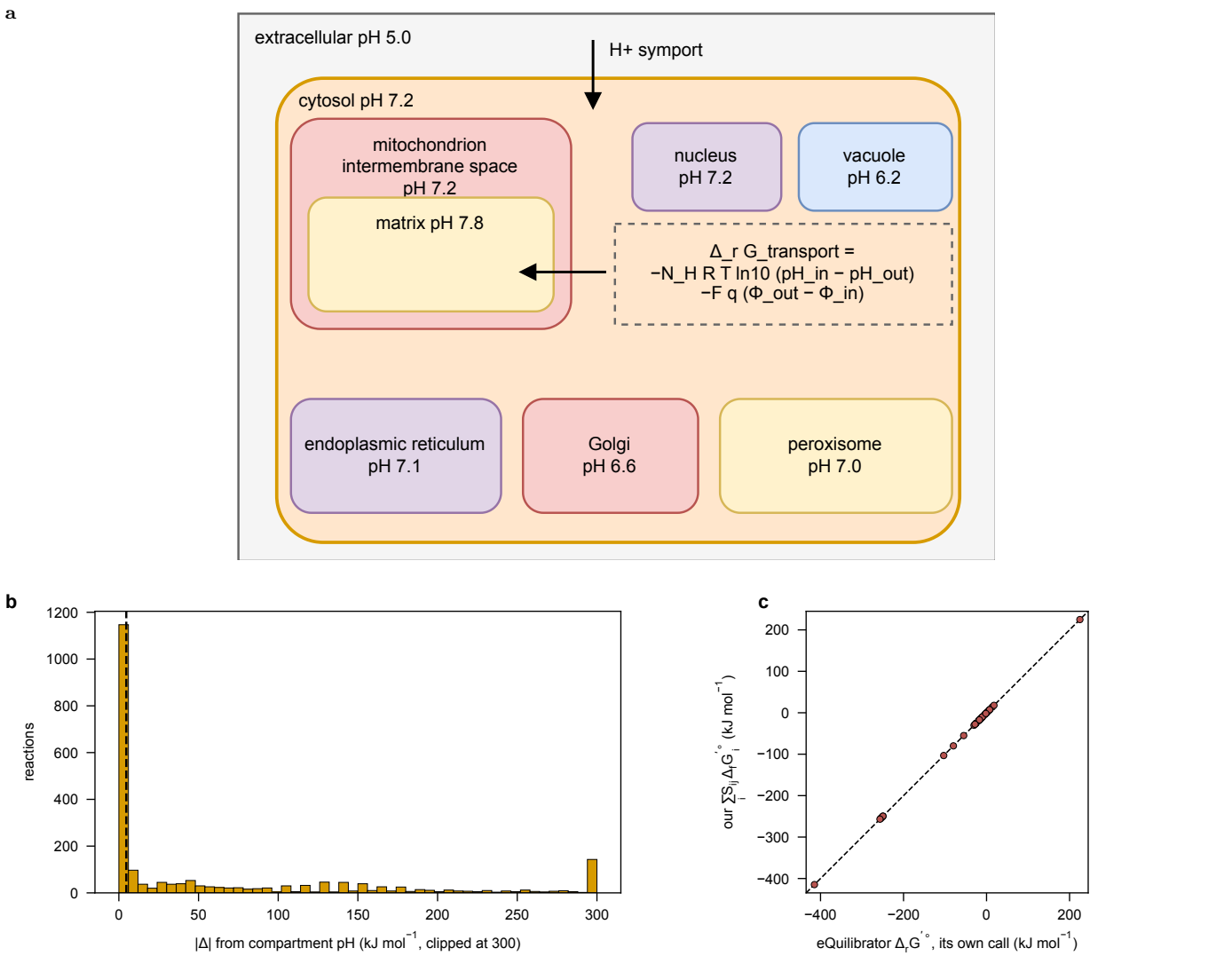


Figure 4: What a compartment is in this pipeline, what applying compartment pH does, and whether the table is assembled correctly. **a**, The compartments yeast-GEM carries, each labeled with the pH this pipeline applies when it transforms a formation energy, and two transport steps: a proton symport across the plasma membrane and a translocation from the cytosol into the mitochondrial matrix, beside the two terms of Equation (1). Every pH here is an assumed textbook value (Table 5). yeast-GEM 9 has no intermembrane-space compartment, so the one drawn takes the cytosolic value the pipeline falls back to. **b**, Absolute change in $\Delta_r G'^{\circ}$ when each metabolite is transformed at its compartment's pH rather than at a uniform 7.0, over the 2,220 reactions that move at all out of the 2,741 covered in both builds, clipped at 300 kJ mol^{-1} for display. Dashed line, the median, 4.67 kJ mol^{-1}; the 90th percentile is 218. The mode at zero is proton-balanced reactions, whose transformed energies have no net pH dependence; the tail is reactions carrying heavily protonated species across a two-unit gradient, where a shift of that size is the transform working rather than failing. **c**, A check of *our assembly*, not of the estimator: reaction energies obtained by summing our per-metabolite table against the same reactions priced by eQuilibrator's own `standard_dg_prime`, over 40 single-compartment mass-balanced reactions, to a maximum absolute difference 9.1×10^{-13} kJ mol^{-1}. The transform and the formation energies are eQuilibrator's in both routes; what is tested is the stoichiometric sum, the metabolite alignment and the zero-proton convention. Generated by `experiments/026-metabolism-flux/scripts/validate_equilibrator_thermo.py` and `plot_equilibrator_thermo.py`; panel a from `notes/assets/drawio/yeast-compartments-ph.drawio`.

3.10 Where the method stops ✗

All five betaxanthin intermediates match their declared molecular formula against a PubChem structure lookup and price from structure alone with no identifier: L-DOPA at −157.68 with $\sigma = 4.30$, dopaquinone at −168.31 with $\sigma = 7.54$, leucodopachrome at −233.01 with $\sigma = 8.79$, betalamic acid at −365.98 with $\sigma = 8.49$ and betanidin at −394.94 with $\sigma = 8.56$ kJ mol^{-1}.

Where it stops is a compound the cache has never seen, since creating one runs a group decomposition that needs pK_a estimates from `cxcalc`. Section 5.4 measures the open-source replacement and Section 5.5 the structures that let it be asked.

Where it stops permanently is a species that is not a molecule, and that is a different statement. A

placeholder formula, a SLIME backbone, a residue inside a macromolecule and a group no training compound carries are each unreachable by any method rather than by this one, and they hold 291 of the model's reactions. Section 5.7 enumerates them and separates them from the 21 that a structure, a ladder or a decomposition would still close.

4 Reproducing Wu et al.'s Fig. 3 ✗

4.1 What the original figure claims ✗

Wu et al. [4] build Yeast-MetaTwin, a comprehensive metabolic network for *S. cerevisiae* obtained by expanding reaction rules over the yeast metabolome, and ask how the resulting *underground* reactions, the promiscuous side activities of enzymes named for a different substrate [5], differ kinetically from the *core* network already in Yeast9 [1]. Their Fig. 3a–h answers it with eight empirical cumulative distributions, one per predictor and parameter: k_{cat} from DLKcat, UniKP, EITLEM-Kinetics, TurNuP and DeepEnzyme, then K_M from Boost_KM, UniKP and EITLEM-Kinetics. Their conclusion is that underground metabolism is separated from known metabolism by K_M rather than by k_{cat}.

The construction is taken from their own notebook (Figure 5). Their split is on whether a reaction identifier contains `rxn`, which is how their tables encode a predicted rather than a curated reaction.

4.2 The per-pair predictions behind Fig. 3 were not released ✗

The authors' own released predictions, split core against underground, are not available.

Their notebook reads its tables from `Results/kcat_km_predict/`, naming `yeast8U_kcat_predict.csv` for DLKcat, `yeast8U_unikp.csv` for UniKP on both parameters, `yeast8U_kcat_result_TurNuP.csv` for TurNuP and `yeast8U_km_km_predict.csv` for Boost_KM. That directory is in neither released artifact. The GitHub repository ships `Results/` with nine subdirectories, none of them `kcat_km_predict`, and its `source_data` holds only `fig2-a`, `fig3-g`, `fig3-j`, `fig4-b`, `fig5-b`, `figs11` and `figS8`; the file named `fig3-g.csv` contains EC-number counts rather than a K_M distribution. The 6.6 GB Zenodo archive (10.5281/zenodo.13911783) contains exactly three top-level directories, `Data`, `Data_retrosynthesis` and `esm`, and of its 1,021,022 entries **zero** match `kcat`, `unikp`, `turnup`, `eitlem` or `deepenzyme`.

The Data Availability statement says data for reproducing all figures is available on GitHub and Zenodo. The **underground arm cannot be drawn**, since generating the Yeast-MetaTwin reaction set is a separate retrobiosynthesis pipeline this work does not rebuild; Figure 5 re-derives the curves by running the same published models over the same organism from hash-pinned mirrors.

4.3 The log base is not a convention ✗

Two of these models exponentiate base 2 and three base 10, and reading either wrong produces turnover numbers that look entirely credible: DLKcat exponentiates with `math.pow(2, ...)`, and reading that as base 10 inflates every turnover number by a power of roughly 3.3.

Table 6: What each predictor was checked against before its output was believed, and the exponentiation base that check established.

Predictor	Checked against	Base
DLKcat	3 released examples, to 4 decimals	$\log_2$
DeepEnzyme	137/137 checkpoint tensors match; 3 independent base checks	$\log_2$
EITLEM	README worked example, 1.3904 vs a stated 1.39	$\log_{10}$
TurNuP	tutorial value, 216.114853 vs 216.114853	$\log_{10}$
EZKinetics	60 rows of the authors' own held-out predictions, median 0.0018 orders of magnitude for k_{cat} and 0.00003 for K_M	$\log_{10}$

Two blockers in an earlier audit also dissolved on inspection. DeepEnzyme's NVIDIA apex requirement is listed in `requirements.txt` and imported nowhere in its code, so it was a training-time mixed-precision dependency and the network runs unmodified. TurNuP's weights exist but are split across two Zenodo records, because the enzyme half needs an ESM-1b fine-tune that ships only with the prediction-function archive.

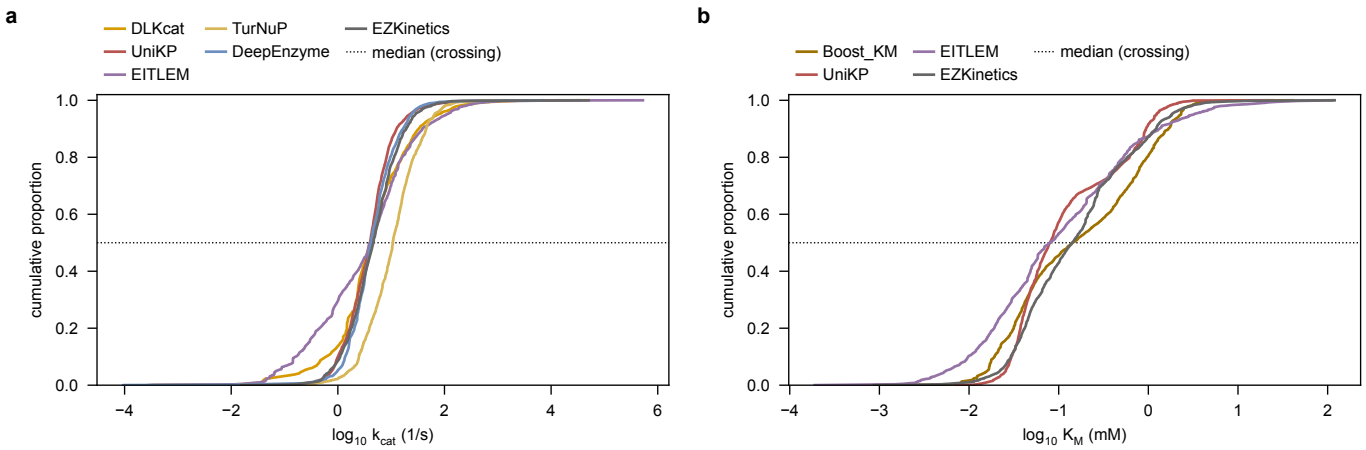


Figure 5: Empirical cumulative distributions of predicted kinetic parameters over yeast-GEM 9.0.2, following the construction of Wu et al. (2026) Fig. 3a–h: values sorted, running proportion plotted. The dotted line at 0.5 is where each curve crosses its own median, which replaces the original's per-curve median tick; six ticks at one height read as marks on the curves rather than as annotations, and one reference line gives every median at once. **a**, the six k_{cat} predictors, overlaid rather than given a panel each, since how far apart the predictors sit is a difference between curves and cannot be read across ten panels. They cross and largely coincide, which is the same fact the per-gene spread of 1.28 orders of magnitude states numerically: a cumulative distribution is invariant to which enzyme got which value. **b**, the four K_M predictors, drawn the same way. EZKinetics is included though it is not one of the original eight. The authors' core and underground curves are absent because the per-pair tables their notebook reads are in neither released artifact. Generated by `experiments/026-metabolism-flux/scripts/plot_kinetic_distributions.py`.

4.4 The comparison between predictors, which the original does not make ✗

Five of the six agree on the median k_{cat} to within 4.0 to 4.6 s^{-1}, and that agreement is superficial. Their *per-pair* rank correlation, on identical enzyme-substrate pairs, ranges from **0.015 to 0.509**.

Table 7: Agreement between predictors on identical enzyme-substrate pairs, in $\log_{10}$ space. The last column is the median absolute difference in orders of magnitude. Produced by `experiments/026-metabolism-flux/scripts/compare_kinetic_predictors.py`.

Pair	n	Pearson	Spearman	Med. |diff|
EITLEM vs UniKP	7,351	0.560	0.509	0.51
TurNuP vs UniKP	5,038	0.458	0.433	0.44
EITLEM vs TurNuP	5,143	0.361	0.330	0.64
DLKcat vs UniKP	7,351	0.337	0.224	0.46
DeepEnzyme vs DLKcat	7,300	0.270	0.284	0.40
DLKcat vs EITLEM	7,351	0.248	0.169	0.70
DLKcat vs TurNuP	5,038	0.142	0.084	0.56
DeepEnzyme vs TurNuP	5,035	0.111	0.094	0.51
DeepEnzyme vs UniKP	7,300	0.109	0.015	0.40
DeepEnzyme vs EITLEM	7,300	0.090	0.025	0.68
K_M: Boost_KM vs UniKP	7,157	0.764	0.789	0.29
K_M: EITLEM vs UniKP	7,351	0.463	0.406	0.50
K_M: Boost_KM vs EITLEM	7,157	0.359	0.345	0.54

That matters for how Fig. 3 should be read. A shift between two curves within one panel is interpreted as a property of underground metabolism, while the spread *between* panels, on the same pairs, is a factor of three to five, so an effect smaller than that sits inside the disagreement between the instruments measuring it.

TurNuP [6] is the exception to the median agreement, at 10.8 s^{-1}, and its coverage is 55.2 percent rather than 95. Both follow from the same property: it consumes the *whole reaction*, and the authors' code invalidates a reaction as soon as one participant lacks a structure. That excludes 1,147 of 2,540 reactions, almost all acyl-chain-resolved lipids.

4.5 The comparison per gene, which is the unit the model reads ✗

Table 7 is keyed by enzyme-substrate pair, and the flux layer reads a different key: a catalytic unit draws one k_{cat} per gene. Each gene's value is the median over its pairs in $\log_{10}$ space.

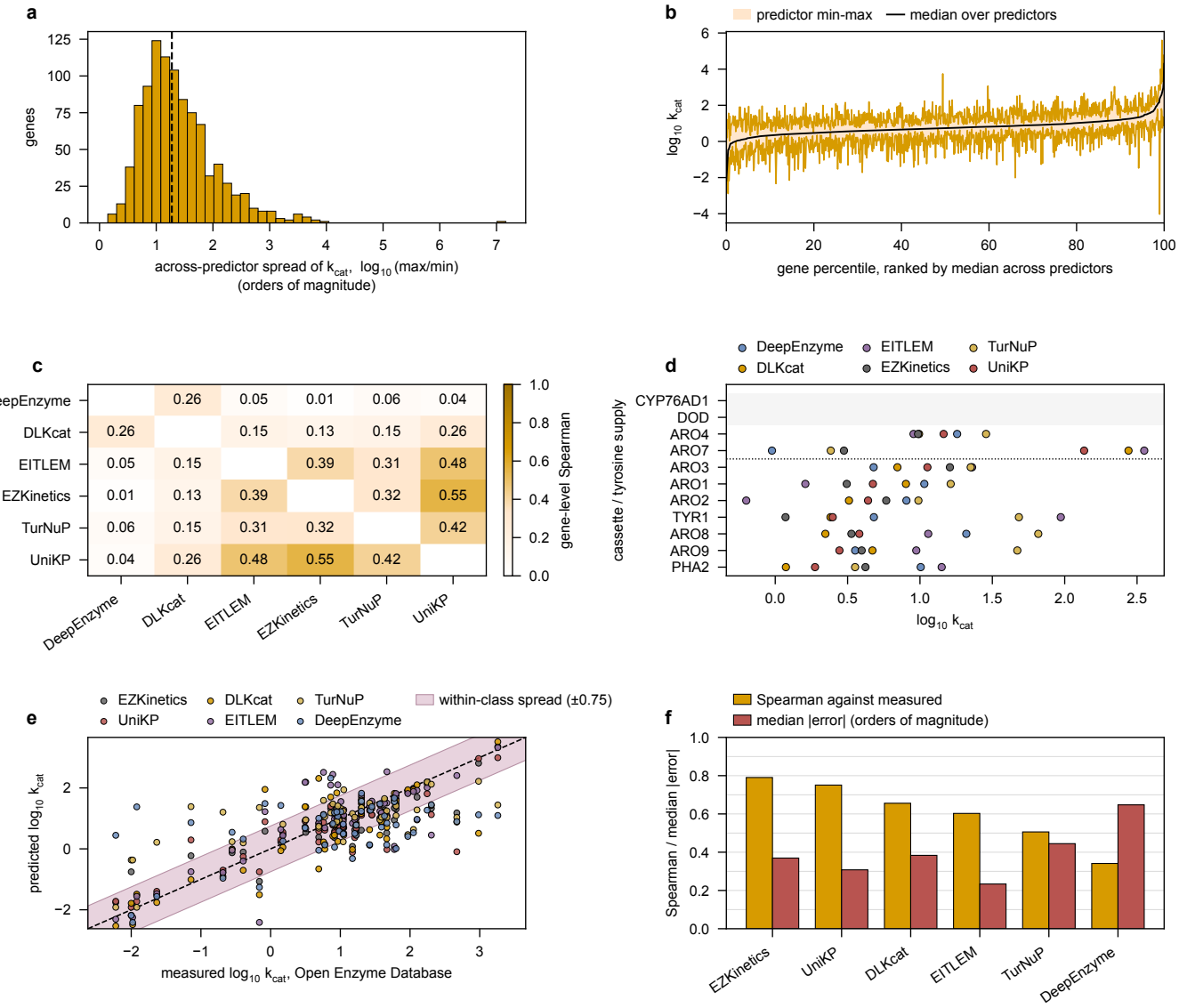


Figure 6: Agreement between the six k_{cat} predictors, and their accuracy against measured yeast values. Panels a to d cover the 980 genes all six predictors cover; e and f the 57 enzyme-substrate pairs that carry a measured value. One color means one predictor throughout. **a**, Across-predictor spread per gene, the difference between the highest and lowest predictor in orders of magnitude; the dashed line is the median, **1.28 orders of magnitude**, a factor of 19, and 71 percent of genes exceed one order of magnitude. **b**, The same spread against the dynamic range of k_{cat} itself, genes ranked by their median across predictors on a percentile axis: the band is the predictor min-max and the black line the median. **c**, Gene-level Spearman between predictors, with the value written in each cell; the diagonal is blank because a predictor against itself is 1 by construction. The pairs run from 0.01 (DeepEnzyme against EZKinetics) to 0.55 (EZKinetics against UniKP). **d**, The genes the betaxanthin case study runs on. Above the dotted rule is the Btx cassette, below it the aromatic-amino-acid genes supplying its tyrosine precursor. The shaded band marks `CYP76AD1` and `DOD`, which are heterologous and have no reaction in yeast-GEM, so no predictor emits a value for them. **e**, Predicted against measured k_{cat} on the 57 pairs the Open Enzyme Database mirror supplies for *S. cerevisiae*, wild-type enzymes assayed between 20 and 40 °C; the dashed line is parity. The shaded band is **biological plausibility, not a tolerance**: its half width of 0.75 is the median standard deviation of $\log_{10} k_{cat}$ across the distinct enzymes of one EC4 class (Section 4.7), so a point inside it is no further from the truth than two enzymes catalyzing the *same* reaction are from each other. Between 56 and 84 percent of each predictor's points fall inside it. **f**, The same comparison ranked, Spearman against median absolute error in orders of magnitude, so the two bars move oppositely for a good predictor. **These accuracy values are an upper bound rather than an estimate**, for the reason Section 4.6 gives. Generated by `experiments/026-metabolism-flux/scripts/plot_per_gene_predictor_agreement.py` from `compare_predictors_to_measured.py`.

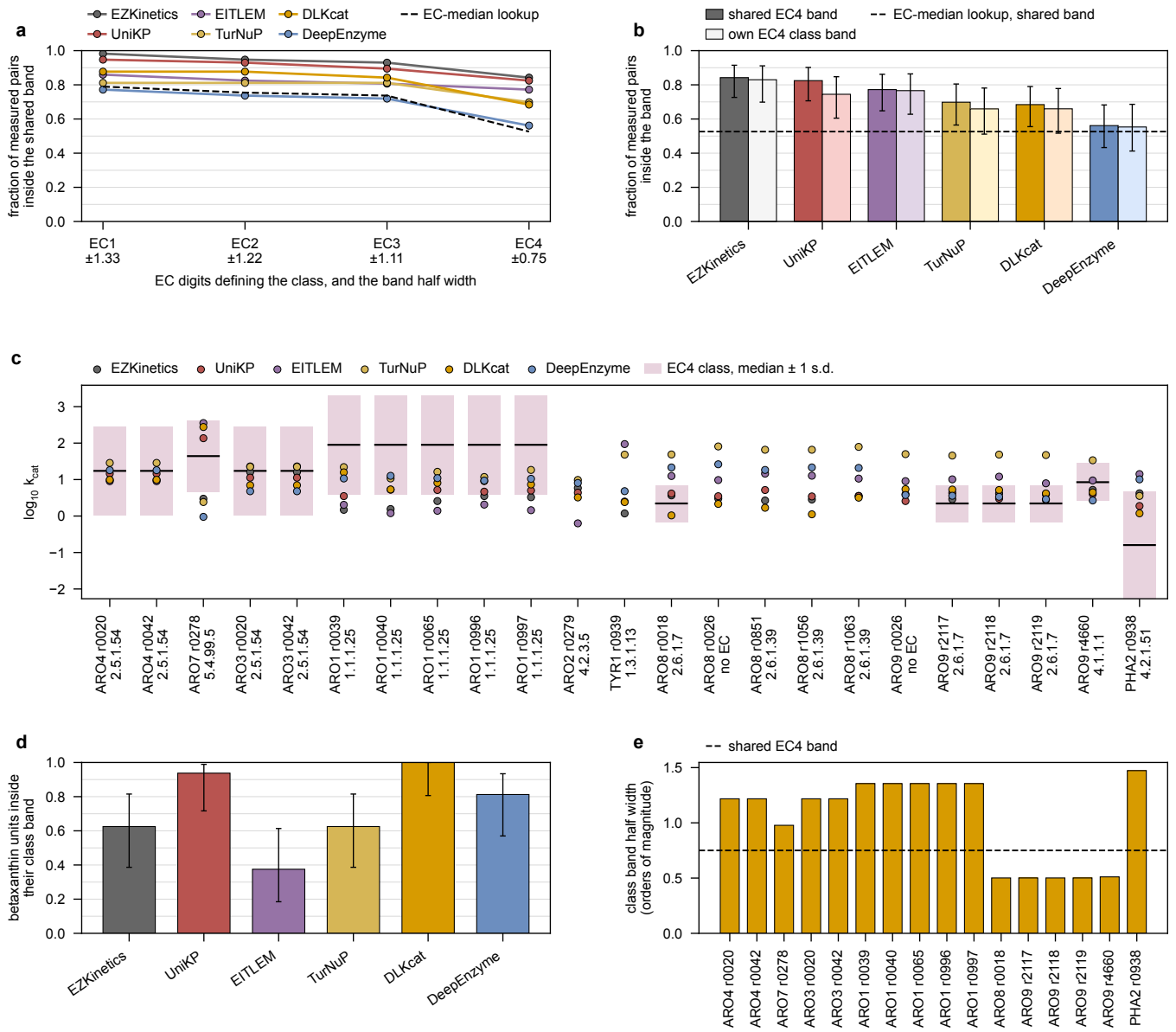


Figure 7: How many of each predictor's k_{cat} values fall inside the spread of enzymes doing the same job. The fraction that Figure 6e gives as a range, made the subject and asked four ways. **a**, Fraction of the 57 measured pairs inside a shared band, as the class is read to fewer EC digits. The band half width at each depth is the median within-class standard deviation over all classes at that depth, 0.75 at EC4 rising to 1.33 at EC1. The dashed line is the EC-median lookup, which predicts each enzyme as the median of the other enzymes sharing its EC number and knows nothing else. At EC4 every predictor beats it, 0.56 to 0.84 against 0.53; at EC1 the lookup reaches 0.79 and the predictors 0.77 to 0.98, so most of what makes a prediction class-plausible is coarse-class knowledge. **b**, The EC4 fraction against the fraction inside each pair's *own* class standard deviation, scorable for the 47 pairs whose class holds at least three measured enzymes; whiskers are Wilson 95 percent intervals. The two readings agree within their intervals for every predictor, EZKinetics 0.84 and 0.83, DeepEnzyme 0.56 and 0.55. **c**, The betaxanthin supply genes, one column per (gene, reaction) unit, 23 units over nine genes, none with a measured yeast value, so class plausibility is the only check available. The muted purple band is one standard deviation about the median of the unit's EC4 class across all organisms, in a hue no predictor uses because it is a reference and not a series; the black tick is the class median and the dots the predictors. A unit with no band has a class with fewer than three measured enzymes; ARO1's five reactions are scored against `1.1.1.25`, the largest of the five classes its pentafunctional annotation carries. **d**, Fraction of the 16 scorable units inside their class band, by predictor. DLKcat 16 of 16, UniKP 15, DeepEnzyme 13, TurNuP and EZKinetics 10, EITLEM 6. EITLEM has the best magnitude accuracy on the measured pairs (Table 8) and runs low here, sitting below the class band on ARO1, ARO4 and ARO3. **e**, The half width of each unit's class band, so d can be read against how wide the target was; the dashed line is the shared EC4 band. The aromatic-pathway classes are 1.2 to 1.4 orders of magnitude wide, so a point inside them is a weak statement. Generated by `experiments/026-metabolism-flux/scripts/plot_within_class_plausibility.py` from `within_class_plausibility.py`.

For k_{cat} the disagreement between predictors is larger than the quantity being predicted, and adding a sixth predictor widened it: spread over consensus range is **1.47**, up from 1.32 over five predictors. **For K_M it is not**, at 0.67 (Figures 6a,b and 8a,b). Both are agreement rather than accuracy, since almost none of these pairs has a measured value, and read that way they arrive at Wu et al.'s conclusion from the other side.

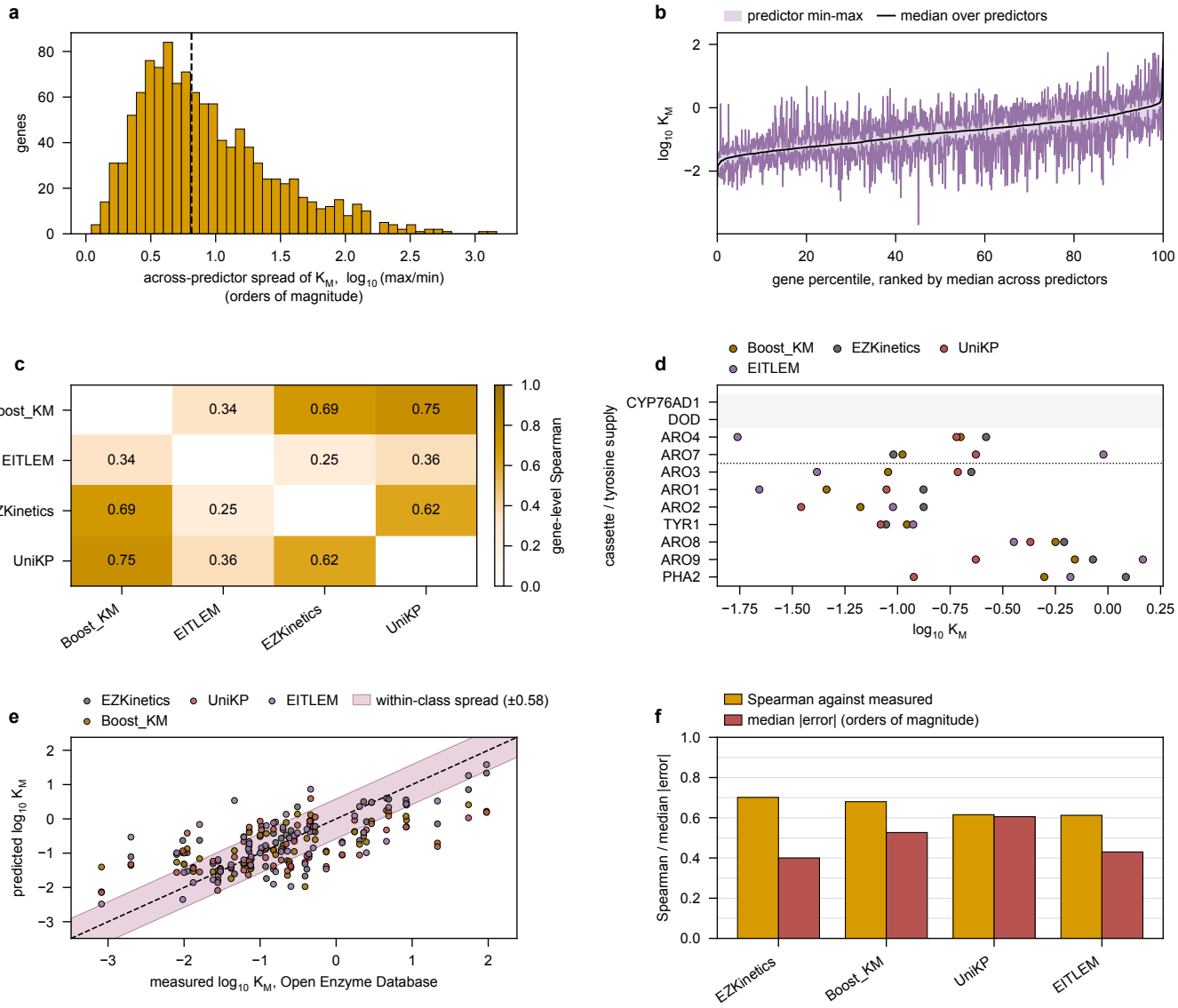


Figure 8: The same six panels for K_M, over the four predictors that emit it. Panels a to d cover the 1,150 genes all four cover; e and f the 57 measured pairs. **a**, The median across-predictor spread is 0.81 orders of magnitude and 36 percent of genes exceed one order of magnitude, against 71 percent for k_{cat}. **b**, The consensus range across genes, 1.22 orders of magnitude from the 10th to the 90th percentile, is wider than the spread, so the ratio is 0.67. **c**, Gene-level Spearman runs from 0.25 (EITLEM against EZKinetics) to 0.75 (Boost_KM against UniKP). **d**, The betaxanthin genes; the shaded band is the two heterologous cassette members no predictor can score. **e**, Predicted against measured K_M on the 57 Open Enzyme Database pairs, carrying the same plausibility band as Figure 6 at ±0.58 orders of magnitude, which is K_M's own median within-EC4-class spread rather than k_{cat}'s. Between 49 and 63 percent of predictions fall inside it, against 56 to 84 percent for k_{cat}, so the four K_M predictors are further from the measurement than the k_{cat} predictors are even after the band is narrowed to match. **f**, The same comparison ranked: EZKinetics leads on rank at +0.70 and EITLEM on magnitude at 0.43 orders of magnitude. The same leakage caveat as Figure 6 applies. Generated by `plot_per_gene_predictor_agreement.py -parameter K_M`.

Class-plausible is not the same as right, and the two parameters separate on it. Every k_{cat} predictor beats the EC-median lookup and no K_M predictor does (Figures 7a and 9a), so on these 57 pairs the K_M predictors add nothing measurable to knowing the enzyme's function.

The consequence for the flux layer. An enzyme-constrained bound $v_j \leq k_{cat,j} E_j$ inherits panel a as its error bar, so a single predictor's table should not be treated as the capacity constraint; the registry of Section 1 exists for that reason, since predictor choice is a hyperparameter with a measured effect size.

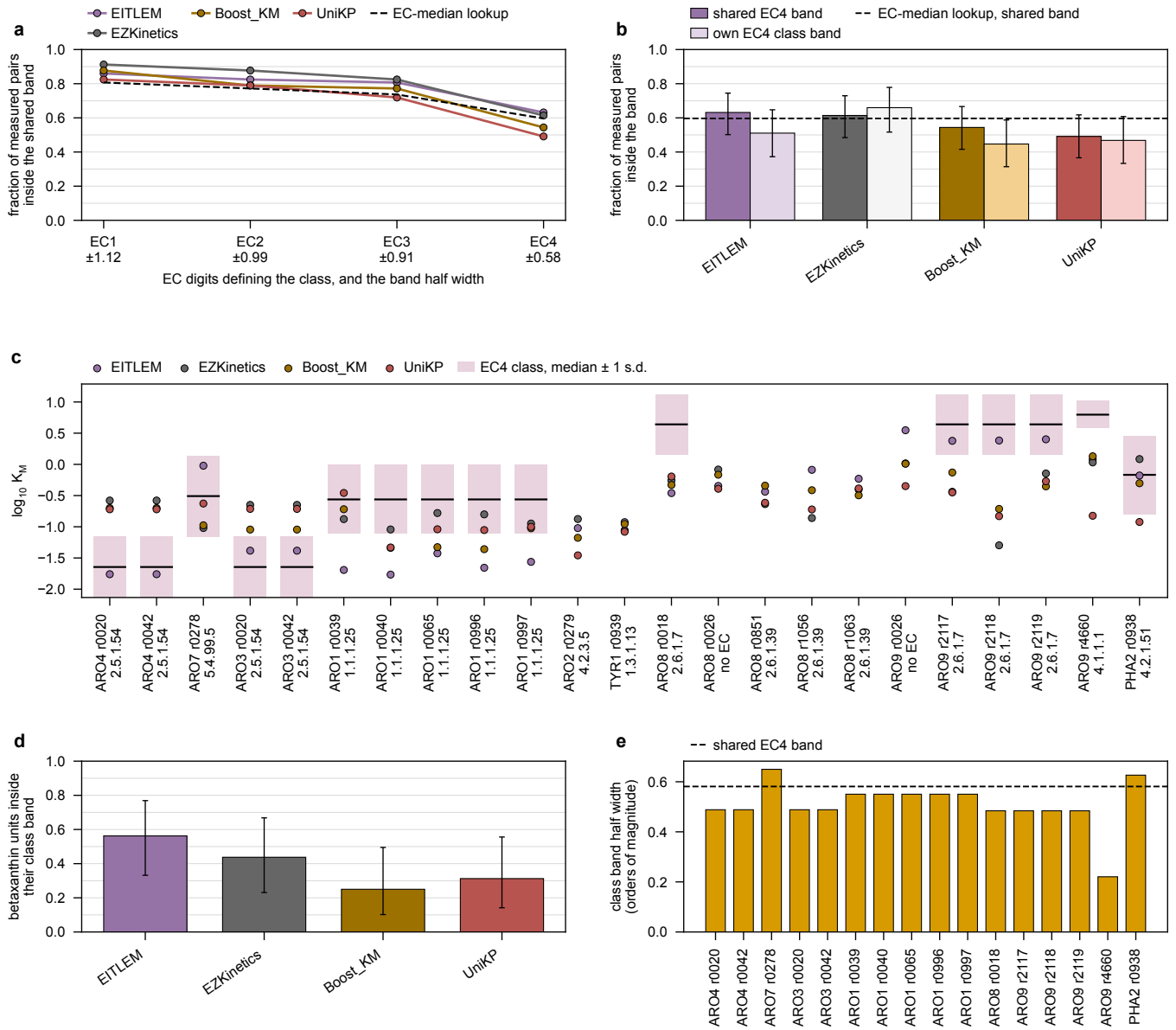


Figure 9: The same five panels for K_M, where the predictors do not beat the lookup. **a**, The shared band is narrower for K_M, 0.58 at EC4 rising to 1.12 at EC1. At EC4 the EC-median lookup reaches 0.60, and no predictor clearly exceeds it: EITLEM 0.63, EZKinetics 0.61, Boost_KM 0.54, UniKP 0.49. **b**, Against each pair's own class band, over the same 47 scorable pairs, EZKinetics reaches 0.66 and the other three 0.45 to 0.51, against 0.49 for the lookup. **c**, The betaxanthin units against their K_M class bands. Most predictions sit outside the band, and on the DAHP synthase units (ARO3, ARO4, `2.5.1.54`) three of the four predictors sit a full order of magnitude above the class. **d**, EITLEM 9 of 16 units inside, EZKinetics 7, UniKP 5, Boost_KM 4. **e**, The K_M class bands are 0.5 to 0.65 wide for every pathway class, so d is a stricter test than its k_{cat} counterpart. Same caveat: plausibility against the function across organisms, not accuracy against the yeast enzyme. Generated by `plot_within_class_plausibility.py -parameter K_M`.

4.6 Accuracy against measured values, and which database can supply them ✗

The Open Enzyme Database mirror carries 1,126 measured *S. cerevisiae* records with a UniProt accession, a substrate, an assay temperature and a pH. Restricting to wild-type enzymes assayed between 20 and 40 °C and joining on (accession, substrate) gives 231 measured pairs, of which **57** also appear among the yeast-GEM catalytic units the predictors were run over.

The ranking is not the agreement ranking. EZKinetics leads on rank accuracy, UniKP is second, and EITLEM leads on magnitude while placing fourth on rank; DeepEnzyme is last on both by a clear margin while sitting mid-pack on agreement. So the pairwise-agreement matrix in panel c cannot be read as a quality ordering. On K_M EZKinetics leads on rank as well, by a smaller margin over Boost_KM, and its bias of -0.05 orders of magnitude is the smallest of any predictor on either parameter apart from EITLEM's.

The leakage caveat is real and is not a reason to discard the numbers. Every one of these predictors trained

Table 8: Accuracy on the 57 measured pairs. Error is the median absolute $\log_{10}$ ratio, so 0.31 orders of magnitude is a factor of 2.0. Bias is the median signed ratio, negative meaning the predictor runs low.

Parameter	Predictor	Spearman	Error	Bias	Within 1 dec
k_{cat}	EZKinetics	**+0.790**	0.37	−0.14	**91.2%**
k_{cat}	UniKP	+0.751	0.31	−0.12	89.5%
k_{cat}	DLKcat	+0.656	0.38	−0.14	84.2%
k_{cat}	EITLEM	+0.603	**0.23**	−0.01	78.9%
k_{cat}	TurNuP	+0.506	0.44	+0.00	81.1%
k_{cat}	DeepEnzyme	+0.341	0.65	−0.23	66.7%
K_M	EZKinetics	**+0.701**	0.40	−0.05	**87.7%**
K_M	Boost_KM	+0.680	0.53	−0.16	78.9%
K_M	UniKP	+0.615	0.61	−0.08	78.9%
K_M	EITLEM	+0.612	**0.43**	+0.02	82.5%

on BRENDA or SABIO-RK, from which the Open Enzyme Database is itself assembled, so these values are an **upper bound** on held-out accuracy. A predictor that cannot reproduce values it may have trained on has a defect no test set forgives, and the ranking between predictors is more robust than any single value because the leakage is common to all.

The measurable subset is not the easy tail, which was worth checking rather than assuming. Accuracy can only be scored on the best-studied enzymes, so the worry is that predictors agree there and the accuracy does not transfer. The across-predictor spread on the 57 pairs is **1.37 orders of magnitude** against **1.27** over all 3,814 pairs, so the scored subset is if anything harder than the model as a whole and the upper-bound caveat stands on leakage alone.

Why the Enzyme Engineering Database supplies no comparison. EnzEngDB [7] was read in full and queried read-only on 2026-09-02. It flattens every assay output into one untyped `fitness_value` column by explicit design, so it holds **no** k_{cat} **and no** K_M; it has no EC field and no organism field; and it contains **zero** occurrences of `yeast`, `cerevisiae` or `Saccharomyces` across the paper, all 140 public experiment records and all 1,342 analysis rows. Its six scaffolds are bacterial and archaeal, and 93 percent of its measurements are whole-cell rather than purified enzyme. Overlap with the 3,728 yeast-GEM catalytic units is **zero**, drawn at zero in Figure 17d. It remains worth mirroring on a different axis, as a sequence-level genotype-to-phenotype corpus of designed variants.

4.7 What that correlation is made of ✗

A Pearson of 0.90 across a quantity spanning six orders of magnitude is not self-evidently good. If most of that range separates enzyme *classes*, a predictor that has learned only which reaction an enzyme catalyzes earns a high correlation while saying nothing about what *this* enzyme's turnover number is, which is what a capacity bound reads. Separating the two cases needs enzymes of the same function, and yeast cannot supply them: its 90 EC4 classes include only four with three or more distinct enzymes. The whole Open Enzyme Database can. It was mirrored under the same provenance rules as the yeast slice, 35,753 records at `sha256 b8d1e652...`, of which 22,975 are wild-type. Every wild-type row carries a k_{cat} *and* a K_M, so the two parameters below are measured on one set of rows rather than on two independent samples.

Table 9: Every scale in the same unit. The first three are properties of the measurements; the fourth is a property of the models. Ranges are max minus min except the total, which is a 10th-to-90th-percentile span because a max minus min over 4,555 enzymes is set by two outliers.

Scale	Decades	Basis
measurement floor	**0.12**	same enzyme and substrate, physiological window, 1,126 repeated pairs
same pair, different conditions	0.40	the 506 of those measured at two or more conditions
within one EC4 class	**1.77**	523 classes holding three or more enzymes
across all enzymes	3.43	4,555 enzymes, 10th to 90th percentile
across predictors, one gene	1.28	six predictors, 980 yeast genes

Enzymes doing the same job differ by far more than the measurement can explain. The median EC4 class spans 1.77 orders of magnitude against a measurement floor of 0.12, a factor of 15 (Table 9). So the within-class spread is biology rather than noise, and it is the floor for any predictor that is really conditioning on function. Three quarters of classes have a standard deviation above 0.5 orders of magnitude and 79 percent span more than a full order of magnitude.

Knowing the exact EC number explains under half the variance. The one-way intraclass correlation rises from 0.04 at EC1 to **0.46** at the full four-digit number for k_{cat} (0.58 for K_M). The intraclass correlation is quoted rather than the R^2, which reaches 0.59 at EC4, because R^2 rises as classes get smaller whether or not the class means genuinely differ, and at EC4 the median class holds four enzymes.

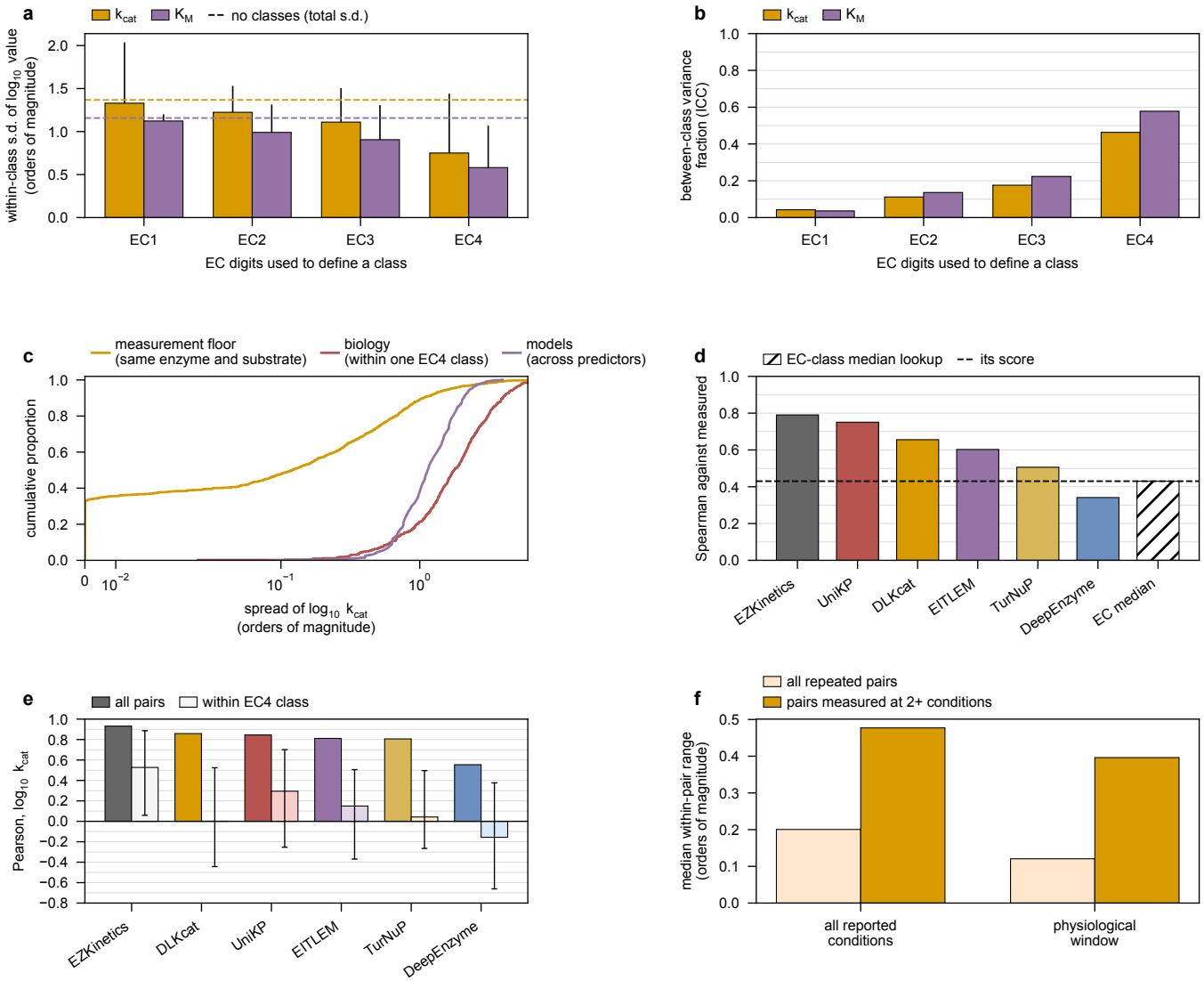


Figure 10: Variation within enzyme function, against the variation a predictor is scored on. a, Within-class spread read at each EC depth: the bar is the median over classes of the standard deviation of $\log_{10}$ value across the distinct enzymes of one class, the whisker its 90th percentile, and the dashed line the spread with no class labels at all. A standard deviation rather than a range, because a range grows with how many enzymes a class happens to have measured. Reading fewer digits pools unlike chemistry, so the spread has to rise, and it does: for k_{cat}, 0.75 orders of magnitude at four digits, 1.11 at three, 1.22 at two, 1.33 at one, against a total spread of 1.37. The three-digit figure is therefore close to no classification at all, and the four-digit number is the only one that describes enzymes catalyzing the same reaction. K_M runs 0.58, 0.91, 0.99, 1.12 against 1.16. **b**, Fraction of variance lying between classes as the EC number is read to more digits, as the one-way random-effects intraclass correlation. **c**, The three spreads as cumulative distributions on one axis rather than as three summary numbers, which is where the separation is visible: the measurement distribution is almost disjoint from the biological one, and the models sit with biology rather than with the measurement. A third of repeated pairs agree exactly, which is the step at zero. **d**, Each predictor against a lookup table that predicts every enzyme as the median of the other enzymes sharing its EC number, leaving out the enzyme itself and backing off to a shorter EC prefix when the class holds no other member (53 of 57 resolve at the full four digits). It uses no sequence, no structure and no learning. **e**, The same correlations before and after centering both axes on the class mean, which removes the between-class range by construction. Only the 27 pairs sharing an EC4 class with another scored pair can contribute, in 13 classes; the interval is a 95 percent cluster bootstrap resampling whole classes. **f**, The measurement floor, for all repeated pairs and for the subset measured at two or more distinct conditions. Generated by `experiments/026-metabolism-flux/scripts/plot_ec_class_variance.py` from `ec_class_variance.py`.

Two predictors do not beat a lookup table. The EC-class median scores a Spearman of **0.430** on the same 57 pairs, with 73.7 percent inside a factor of ten. DeepEnzyme sits below it and TurNuP barely above, while EZKinetics and UniKP clear it by a margin a class lookup cannot explain (Table 8).

Table 10: Pearson on $\log_{10} k_{cat}$ over the 27 pairs that share an EC4 class with another scored pair, before and after both axes are centered on their class mean. The interval is a 95 percent cluster bootstrap over the 13 classes. **At this sample size only EZKinetics is resolved away from zero, and no two predictors are resolved from each other.**

Predictor	All pairs	Within class	95% interval
EZKinetics	+0.931	**+0.527**	[+0.06, +0.89]
DLKcat	+0.858	+0.001	[−0.44, +0.52]
UniKP	+0.845	+0.294	[−0.25, +0.70]
EITLEM	+0.811	+0.149	[−0.37, +0.51]
TurNuP	+0.806	+0.042	[−0.27, +0.50]
DeepEnzyme	+0.554	−0.156	[−0.66, +0.38]

The correlation is mostly the class. Every predictor loses most of it when the between-class range is removed. Read with the lookup-table result, these models rank enzyme *functions* much better than they rank enzymes *within* a function, and it is the second that an enzyme-constrained bound needs, since a flux model already knows which reaction it is parameterizing. What is established is the size of the within-class signal rather than an ordering on it.

4.8 EZKinetics, the seventh predictor and the only condition-aware one ✗

EZKinetics (Le Yuan, Zhengyi Zhang and Huimin Zhao) was added after the five above. It is **unpublished**, with the manuscript under review, so it has no DOI and is pinned by repository and commit (`le-yuan/EZKinetics`, MIT since 2026-04-16, revision `bf7a897`) rather than cited. It is also the only predictor here that ships its weights inside the repository rather than on figshare, Zenodo or Hugging Face.

Which of its 36 models to run is not a matter of taste. The repository holds three parameters crossed with four protein encoders and three substrate encoders, and any pair produces numbers. `inference/predict.py` fixes the combination, and the shipped scalers confirm it independently through their feature widths: k_{cat} uses ESM2_650M with ChemBERTa-77M-MTR, $1280 + 384 + 2 = 1666$, and K_M uses ESM2_3B, $2560 + 384 + 2 = 2946$. The two extra features are pH and temperature. Output is base 10, the same convention as UniKP.

Table 11: Reproducing the authors' own held-out predictions, over 60 rows of each parameter's test split, as absolute $\log_{10}$ ratio against their shipped `y_pred`. Each split carries the assay conditions and aligns row for row with the shipped predictions, verified on all 6,676 k_{cat} rows.

Parameter	Conditions passed	Median	p90	Max
k_{cat}	true assay pH and temperature	**0.0018**	0.032	0.077
k_{cat}	the 7.0 / 25.0 defaults	0.046	–	0.766
K_M	true assay pH and temperature	**0.00003**	0.023	0.102
K_M	the 7.0 / 25.0 defaults	0.024	–	0.300

The reproduction gate is on the median, and that is a deliberate choice. A max-based gate would be the wrong statistic: reproduction here is not bit-exact and cannot be, since our ESM2 forward pass runs on different hardware and a different torch build. The residual survives setting the protein batch size to 1, so it is not attention over padding; beyond that the cause is not established and no explanation is offered. What is established is the scale, and 0.03 orders of magnitude at the 90th percentile against an across-predictor disagreement of 1.28 orders of magnitude cannot move any conclusion here.

Conditioning is worth a median 0.046 orders of magnitude, and we cannot use it, as much as 0.77 on a single row. yeast-GEM carries no per-pair assay pH or temperature, so **the production run is necessarily condition-blind**, at 7.0 and 25.0 °C, which keeps EZKinetics comparable with the six condition-blind predictors and means its Table 8 accuracy is measured with one input held at a default. It leads both rankings anyway. The K_M table was built the same way, with the ESM2_3B encoder its model expects, after the same gate passed at a median of 0.00003 orders of magnitude.

Coverage is 3,552 of 3,728 catalytic units, the same substrate-structure ceiling DLKcat, UniKP and EITLEM hit; median predicted turnover is $4.62\,s^{-1}$, inside the 3.98 to 4.17 band the other sequence-plus-SMILES predictors occupy.

5 Coverage, route by route ✗

Every number in this section is produced by a script in `experiments/026-metabolism-flux/`, named in the source note of each figure and table; `thermo_pipeline_yeast9.py`, `thermo_curation_price.py` and `external_pka_`

`apply_selection.py` need the eQuilibrator env rather than the torchcell env.

A value with no named route is not a value. Every one of yeast-GEM 9.0.2's 2,806 metabolites carries either a *route*, which says how its formation energy was produced, or an *unpriced reason*, which says what stands in the way; the same holds for each of the 4,131 reactions. The two partition the model exactly, and the partition is asserted when the table is written rather than checked afterwards. This section reads that partition out, route by route.

Table 12: Every term the recovery sections use, in alphabetical order. A route is a way of producing a value, a failure class is a reason there is none, and the two partition every species and every reaction in the model.

Term	Definition
admission	the strongest claim a curation record's sources support: measured, tabulated, computed or provisional. Only measured and tabulated reach a served table
cache, by accession	a route. Reads an eQuilibrator compound the model's own identifier resolves, takes its component-contribution formation energy and transforms it to the metabolite's compartment
cache, by written structure	a route. Writes a structure from the metabolite's name, hashes it to an InChIKey and finds a compound eQuilibrator already holds under that key
cancellation	a route. Drops a species appearing unchanged on both sides, so a reaction closes without that species ever being priced
carrier potential	a route. Gauges an electron carrier pair, oxidized form at zero and reduced form at $-nFE'^{\circ}$, so the arbitrary gauge cancels in any reaction that conserves the carrier
catalytic unit	one reaction and the gene product complex that catalyzes it, the key a turnover number and a Michaelis constant are stored under
CODATA ion table	a route. Reads a tabulated single-ion formation energy, shifts it to eQuilibrator's gauge and applies the Debye-Huckel term
component contribution	eQuilibrator's estimator: decompose a structure into groups, fit group energies against a measured reaction corpus, and return a formation energy with the covariance that fit implies
composition surrogate	a route. Transfers the energy of a cache compound sharing the heavy-atom composition, admitted only where every such compound agrees to 5 kJ/mol
conserved moiety	a route. Decomposes every participant leniently and prices the group difference where the atoms no group covers match as a multiset across the two sides
curated structure or convention	a route. Prices a species from a structure or a polymer convention a curation record fixes, with sources and a sha256 per source
decomposition refused	a failure class. the structure exists and component contribution refuses it, so a pKa predictor changes nothing
evidence class	what a value rests on, one of measured, tabulated, estimated by component contribution, or curated
external pKa cascade	a route. Creates a compound eQuilibrator has never seen, from the pH 7 major microspecies plus a predicted pKa ladder, then decomposes and prices it
flux-carrying	a reaction whose flux-variability width is nonzero under the wild-type medium, so the layer can express a flux through it
group in the null space	a failure class. the structure decomposes and one group is in the component-contribution null space, so the estimate has infinite variance
microspecies	one protonation state of a compound. The reference microspecies is the state the caller draws, and choosing the neutral parent rather than the pH 7 major microspecies moves a formation energy by hundreds of kJ/mol
mixed routes	a route. Sums participants priced by more than one route, so the reaction's uncertainty belongs to no single route
more than one reason	a failure class. the reaction is blocked by species in two or more of the classes above
no structure	a failure class. a definite formula with no structure in any namespace and no writer that composes one from the name
non-chemical reaction	a failure class. an exchange, demand, sink or biomass pseudoreaction, not elementally balanced, so a standard reaction energy has no meaning for it
permanently closed	a reaction no pKa predictor, structure writer, decomposer or literature table can price, because every species blocking it is not a molecule
pKa ladder	the list of macroscopic dissociation constants inside pH 0 to 14 that fixes how a compound's protonation states are populated, and therefore its Legendre transform
placeholder formula	a failure class. an R, X or $*$ in the formula: the species is a family of molecules and no single formation energy exists for it
placeholder structure	a failure class. the only structure that resolves carries an attachment point, so the species is a residue inside a macromolecule
priced	carrying a transformed formation energy at the species' own compartment conditions; a reaction is priced when every participant is, or when a reaction-level route fills it directly
pseudo-metabolite	a failure class. a modeling device, a pool, a SLIME backbone or a difference formula, rather than a molecule
recorded, not admitted	a failure class. a curation record exists and its admission is computed or provisional, so it is reported and not priced
repeat-unit surrogate	a route. Prices one polymer residue as the monomer minus water minus half a calibrated hydrolysis energy
shipped metabolite column	a route. Reads yeast-GEM's own `model_metDeltaG.csv`, converted out of kilocalories at parse time
shipped reaction column	a route. Reads yeast-GEM's `model_rxnDeltaG.csv` directly
target	one distinct structure submitted to the external pKa route. Several yeast-GEM species can share one target

5.1 What the tables cover ✗

Table 13: Every source of a formation energy counted against the whole model, 4,131 reactions and 2,806 metabolites. A reaction counts only when every participant carries a value, which is the level the flux layer consumes, since the second-law hinge needs the whole stoichiometric sum; the metabolite column sits beside it because a metabolite-level figure is the one usually quoted and it overstates what is usable. The last row is the pipeline as built, which reaches every source above and eight recovery routes besides (Table 14). The shipped row's values are on the wrong scale, being yETFL's in kcal mol^{-1} and labeled kilojoules (Section 3.4), so its count is a count of numbers rather than of reactions a consumer can read without a conversion.

Source	Rxns	%	Mets	%	Supplies
yeast-GEM shipped column	3,505	84.85	2,389	85.14	kcal-scale point values, no uncertainty, this model only
eQuilibrator recomputation, baseline	2,741	66.35	2,113	75.30	covariance of rank 590, any organism, any compound with a structure
union of the two	3,716	89.95	2,511	89.49	covariance where the recomputation reaches, point values elsewhere
union plus the tight surrogate	3,730	90.29	2,522	89.88	as the union, plus 39 species a same-composition surrogate determines
union plus every surrogate	3,862	93.49	2,574	91.73	refused: the median surrogate spread is 52.6 kJ/mol, which orients nothing
as built	**3,801**	**92.01**	**2,559**	**91.20**	one covariance-carrying estimate wherever one exists, and a named route and provenance record on every value

Over the 2,845 reactions with nonzero flux-variability width under the wild-type medium, the last row is 2,724, 95.75 percent.

3,801 of 4,131 reactions carry a standard reaction energy, and 2,724 of the 2,845 that can carry flux. Counted over chemical reactions alone, dropping the 284 exchange, demand, sink and biomass pseudoreactions for which a standard reaction energy has no meaning, it is 3,535 of 3,847. Those three denominators answer three different questions and none of them is the headline: the flux-carrying figure bounds what the layer can express, the chemical figure is what a thermodynamics claim should quote, and the whole-model figure is what a reader compares against yeast-GEM's shipped column.

The ladder's first four rows are a comparison of sources and the last is the pipeline. Reading the recomputation's 66.4 percent against the shipped column's 84.9 was never the right comparison, because the two price different reactions rather than nested ones and only one of them carries an uncertainty. Three things separate them.

Covariance. The recomputation returns a square-root factor of the formation-energy error covariance, of rank 590 over 2,806 metabolites, so a reaction energy comes with an uncertainty correlated with every other reaction's in the way component contribution says it should be. The shipped file is a column of numbers with no uncertainty of any kind. Of the reactions the built table prices, 1,770 carry a finite standard error, which is the stricter figure and the one a sampler would use.

Transferability. The shipped table was assembled by matching yETFL [8] values on the metabolite name string, so it exists for this model and no other; the recomputation works from structure and prices all five betaxanthin intermediates (Section 3).

Transport terms. **Neither table carries one, and the module now computes it**, nonzero on 326 of the 1,468 reactions that cross a membrane and agreeing in sign with the GEM's own column on 84.7 percent of the 287 nonzero in both (Section 3.9, `results/transport_terms.json`). **This is the one row where the shipped table has a partial answer the recomputation cannot reconstruct**, since yETFL's term carries a protonation contribution this module does not.

5.2 Every value, and the route that produced it ✗

The cache is the workhorse and the recovery routes are the tail, and the tail is where the errors live. 2,117 metabolites come from an identifier that resolved to a compound eQuilibrator already holds, and a further 289 from a structure written here whose InChIKey resolved one. The remaining 153 come from seven routes each worth between 2 and 70 species, and each of those seven had to be built, checked and given a provenance record. Their reaction yield is smaller still: 19 reactions carry the cancellation route, 9 the external pK_a cascade, 2 a curated structure and 1 a conserved moiety. **Coverage was not what the recovery routes bought.** What they bought is a table in which every value names its own method, so a disagreement can be attributed rather than argued about.

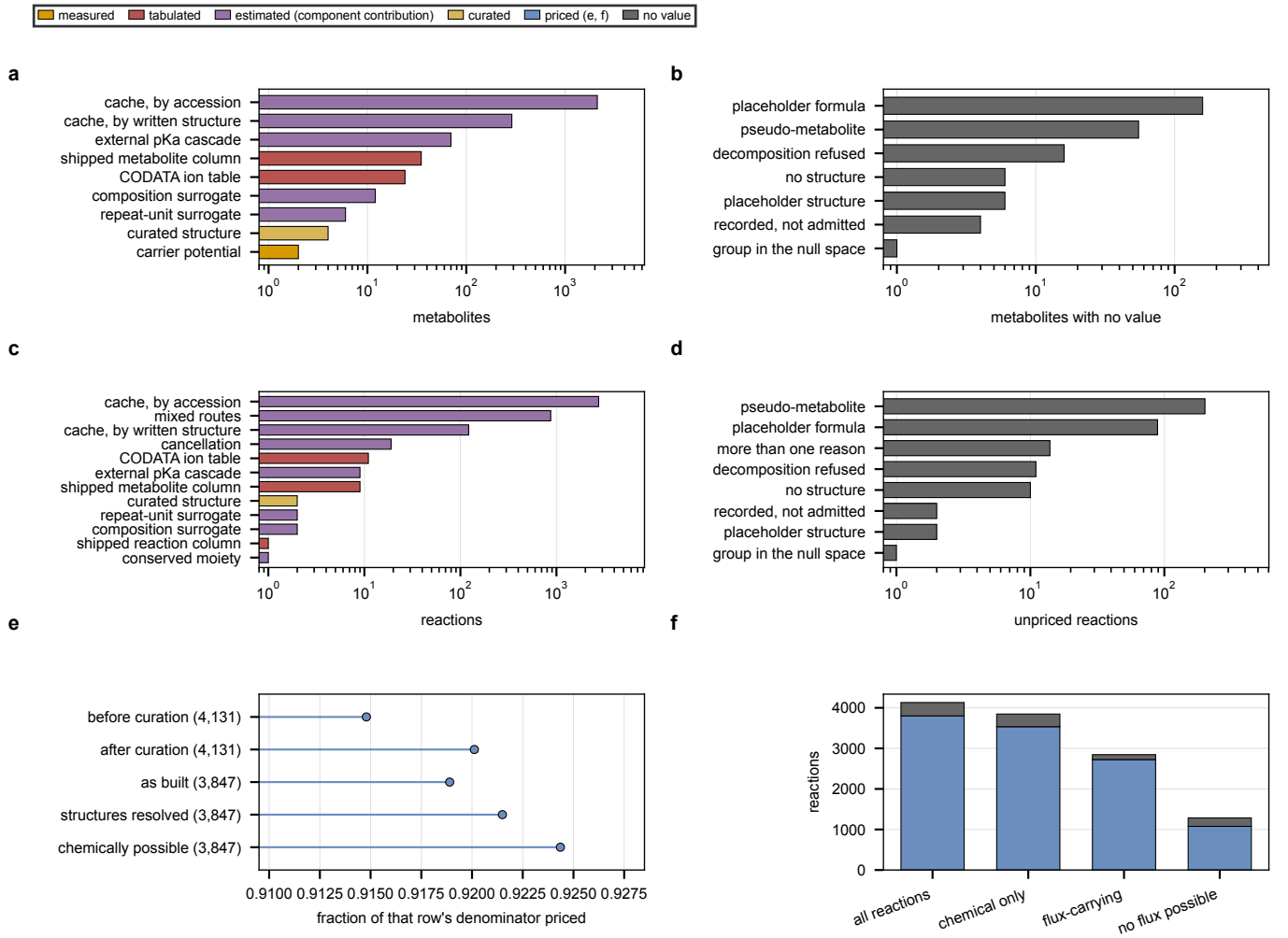


Figure 11: Which route produced each formation and reaction energy, and which failure class holds the rest. Bars are colored by evidence class, and panels a to d use a logarithmic count axis because the routes span three orders of magnitude. **a**, The 2,559 priced metabolites by route. Two routes reach the eQuilibrator compound cache, one by an identifier the model carries and one by a structure written here from the metabolite's name, and together they account for 2,406 of them. **b**, The 247 metabolites with no value, by failure class. The two largest classes are species that are not molecules. **c**, The 3,801 priced reactions by route. *Mixed routes* is a reaction whose participants came from more than one route, so its uncertainty belongs to no single one, and it is the second largest entry. **d**, The 330 unpriced reactions by the single failure class that blocks them, which partitions them exactly; *more than one reason* is the class for a reaction blocked by species in two or more of the others. **e**, The ceiling ladder. Each row is a fraction of the denominator printed in its own tick label, because the before-and-after-curation pair exists only over the whole model while the three rungs are over the 3,847 chemical reactions; the run that produced the before figure did not emit a chemical denominator. Curation moved the whole model from 3,779 to 3,801 reactions. The last rung, 3,556, is what would be priced if every structure, ladder and refused decomposition in the model were resolved, so it is the ceiling for any method rather than for this one. **f**, Reactions priced under four denominators. The reactions no medium lets carry flux hold most of the remaining gap, so the flux-carrying restriction is higher than the whole-model figure rather than lower. Generated by `experiments/026-metabolism-flux/scripts/plot_thermo_recovery_routes.py`.

5.3 Mapping a model species onto a molecule ✗

Four mappings sit between the model and an estimator, and each can be silently wrong: an identifier can resolve to the wrong compound, a name can be parsed into the wrong structure, a polymer's residue can be counted with the wrong end groups, and a curator can record a species the model does not actually contain.

The accession route had no composition check, and eight species were priced as a different molecule. It accepted the first identifier that resolved to a compound carrying a formation energy, whatever molecule that compound is. Checking the resolved structure's composition against yeast-GEM's own formula, over every accession-priced species, separates three populations (Figure 12b). **Six prices are withdrawn and the withdrawal costs 10 reactions**, which is the correct direction: a formation energy for the wrong molecule is worse than none, because it produces a reaction potential with a confident sign.

The 12 dolichols are kept, with the offset recorded. Every dolichol species in yeast-GEM carries the $n = 0$ core

Table 14: Every route that produced a formation energy, and how many of yeast-GEM 9.0.2's 2,806 metabolites it priced. The nine routes sum to the 2,559 priced species; the 247 with no value are Table 22. Evidence class is the strongest thing the number rests on, so a surrogate is estimated rather than tabulated: its value is component contribution on another molecule.

Route	What it does	What it needs	Evidence	Species
cache, by accession	reads an eQuilibrator compound the model's own identifier resolves, takes its component-contribution formation energy and transforms it to the metabolite's compartment	a MetaNetX, ChEBI, KEGG or BiGG identifier on the metabolite	estimated	2,117
cache, by written structure	writes a structure from the metabolite's name, hashes it to an InChIKey and finds a compound eQuilibrator already holds under that key	a name stating the acyl composition, and a head-group template	estimated	289
external pKa cascade	creates a compound eQuilibrator has never seen, from the pH 7 major microspecies plus a predicted pKa ladder, then decomposes and prices it	a structure and a pKa ladder from the selected cascade	estimated	70
shipped metabolite column	reads yeast-GEM's own `model_metDeltaG.csv`, converted out of kilocalories at parse time	a row in that file for this metabolite	tabulated	35
CODATA ion table	reads a tabulated single-ion formation energy, shifts it to eQuilibrator's gauge and applies the Debye-Huckel term	a monatomic ion whose species CODATA evaluates	tabulated	24
composition surrogate	transfers the energy of a cache compound sharing the heavy-atom composition, admitted only where every such compound agrees to 5 kJ/mol	a definite formula the cache can decompose elsewhere	estimated	12
repeat-unit surrogate	prices one polymer residue as the monomer minus water minus half a calibrated hydrolysis energy	a hydrolysis reaction in the model that names the monomer	estimated	6
curated structure or convention	prices a species from a structure or a polymer convention a curation record fixes, with sources and a sha256 per source	an admitted curation record of kind structure or polymer convention	curated	4
carrier potential	gauges an electron carrier pair, oxidized form at zero and reduced form at $-nFE'^{\circ}$, so the arbitrary gauge cancels in any reaction that conserves the carrier	a measured midpoint potential for the couple	measured	2

formula while every accession structure is the $n = 1$ molecule, so the offset is one isoprene unit on both sides of any reaction that conserves the carrier and cancels there. That is a statement about the reactions these species appear in rather than about the values, so the offset is written into the provenance record rather than corrected away.

A curation record is a claim with sources, and most of them close nothing. Each of the 16 records carries its own structure, potential, pK_a ladder or polymer convention, with a verbatim quote, a retrieval command and a sha256 for every source, and each records what the group decomposer said about it. Eleven are admitted and five are not: three are provisional, resting on another organism or on a construction rather than a measurement, and two record that nothing was found. **An admitted record still prices nothing when the structure it records is one component contribution refuses**, which is four of the nine tabulated records and is the honest outcome rather than a failure of the curation.

5.4 A pK_a route that does not need the license ✗

Coverage of the native network and coverage of an inserted pathway are separate problems. A perturbation understood as an edit to the cell's total genomic content adds reactions yeast-GEM never had, carrying compounds absent from the compound cache, which is the one case the ChemAxon license governs. Adding such a compound requires the pK_a constants `cxcalc` supplies, without which `LocalCompoundCache` opens read-only; that blocker also stands behind every metabolite the structure writers reach and the cache does not already hold.

The bridge is an unreleased option, and it takes pK_as and nothing else. `equilibrator-assets` master merged `pkas_provided_externally` on 2026-08-15; the last PyPI release, 0.6.0 from January 2024, does not carry it, so the route runs from a clone pinned at commit `c79e371e`. The caller passes a dictionary keyed by the exact structure string, valued by a list of *macroscopic* constants, and the option lifts read-only mode without ChemAxon. Charge and the proton ladder are then derived from the structure rather than supplied: the atom bag comes from the SMILES through Open Babel, and the microspecies mapping is indexed off the hydrogen count and the number

Table 15: Every route that produced a reaction energy, over yeast-GEM 9.0.2's 4,131 reactions, 3,801 of them priced. A reaction is priced only when every participant carries a formation energy, or when a reaction-level route fills it directly, so the largest entry is the metabolite route its participants all came from and the second is the reactions assembled from more than one.

Route	What it does	What it needs	Evidence	Reactions
cache, by accession	reads an eQuilibrator compound the model's own identifier resolves, takes its component-contribution formation energy and transforms it to the metabolite's compartment	a MetaNetX, ChEBI, KEGG or BiGG identifier on the metabolite	estimated	2,751
mixed routes	sums participants priced by more than one route, so the reaction's uncertainty belongs to no single route	every participant priced by something	estimated	872
cache, by written structure	writes a structure from the metabolite's name, hashes it to an InChIKey and finds a compound eQuilibrator already holds under that key	a name stating the acyl composition, and a headgroup template	estimated	122
cancellation	drops a species appearing unchanged on both sides, so a reaction closes without that species ever being priced	an unpriced species identical in name, formula and charge on both sides	estimated	19
CODATA ion table	reads a tabulated single-ion formation energy, shifts it to eQuilibrator's gauge and applies the Debye-Huckel term	a monatomic ion whose species CODATA evaluates	tabulated	11
external pKa cascade	creates a compound eQuilibrator has never seen, from the pH 7 major microspecies plus a predicted pKa ladder, then decomposes and prices it	a structure and a pKa ladder from the selected cascade	estimated	9
shipped metabolite column	reads yeast-GEM's own `model_metDeltaG.csv`, converted out of kilocalories at parse time	a row in that file for this metabolite	tabulated	9
curated structure or convention	prices a species from a structure or a polymer convention a curation record fixes, with sources and a sha256 per source	an admitted curation record of kind structure or polymer convention	curated	2
repeat-unit surrogate	prices one polymer residue as the monomer minus water minus half a calibrated hydrolysis energy	a hydrolysis reaction in the model that names the monomer	estimated	2
composition surrogate	transfers the energy of a cache compound sharing the heavy-atom composition, admitted only where every such compound agrees to 5 kJ/mol	a definite formula the cache can decompose elsewhere	estimated	2
shipped reaction column	reads yeast-GEM's `model_rxnDeltaG.csv` directly	a row in that file, with the 33 dGPredictor rows exempt from the conversion	tabulated	1
conserved moiety	decomposes every participant leniently and prices the group difference where the atoms no group covers match as a multiset across the two sides	a curated structure for each participant and a moiety that cancels	estimated	1

Table 16: Which committed script documents and produces each route. A route present in `results/thermo\_pipeline\_yeast9.json` with no entry here makes `thermo\_route\_tables\_tex.py` raise, so the table cannot fall behind the pipeline.

Route	Produced and documented by
cache, by accession	`build_equilibrator_thermo.py, thermo_accession_sweep.py`
cache, by written structure	`lipid_structure_writer.py`
external pKa cascade	`external_pka_apply_selection.py`
shipped metabolite column	`yeast9_delta_g_provenance.py`
CODATA ion table	`fetch_ion_formation_energies.py, thermo_ion_gauge_check.py`
composition surrogate	`improve_parameter_coverage.py`
repeat-unit surrogate	`thermo_polymer_repeat_unit.py`
curated structure or convention	`thermo_curation_price.py, results/curation/`
carrier potential	`thermo_curation_price.py, results/curation/cytochrome_c.json`
mixed routes	`thermo_pipeline_yeast9.py`
cancellation	`thermo_pipeline_yeast9.py`
shipped reaction column	`thermo_dgpredictor_rows.py`
conserved moiety	`thermo_curation_price.py`

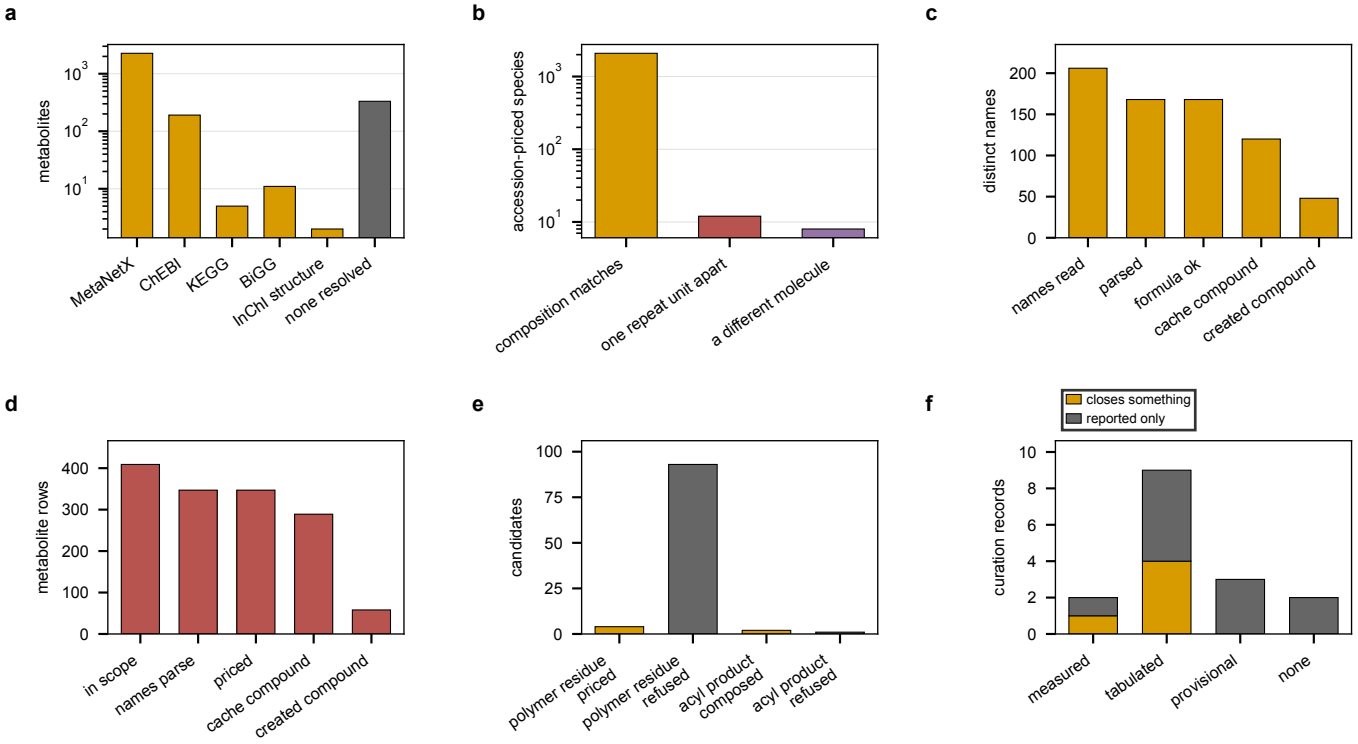


Figure 12: The four mappings between a yeast-GEM species and a molecule an estimator can price. a, Which namespace resolved each metabolite, in the order `build_equilibrator_thermo.py` tries them, so the bars are disjoint: a metabolite counted under ChEBI is one whose MetaNetX identifier missed. MetaNetX 2,265, ChEBI 191, BiGG 11, KEGG 5 and an InChI structure lookup 2, with 332 resolving nothing. **b**, The composition check over the 2,414 accession-priced species: whether the compound the identifier resolved carries the model's own heavy-atom composition. 2,087 match, 12 are the dolichol family whose structures carry exactly one isoprene unit more than the model's formula, and 8 are a different molecule. **c**, The lipid structure writer counted in distinct names. 168 of 206 parse, all 168 reproduce yeast-GEM's own elemental formula, 120 resolve a compound the cache already holds and 48 need a compound created through the external pK_a route. **d**, The same writer counted in metabolite rows, which is the count coverage moves by: 347 of 409 rows priced, 289 by cache lookup and 58 through a created compound. **e**, The two smaller writers. The polymer repeat-unit surrogate prices 4 of 97 candidates, refusing the rest because no hydrolysis reaction in the model identifies a monomer; the acyl-transfer writer composes 2 of 3 products and prices none of them, pricing being the pK_a route's job. **f**, The 16 curation records by admission, split by whether the record closed a species or a reaction. Only *measured* and *tabulated* may price anything, and four of the nine tabulated records still close nothing because the structure they record is one component contribution refuses. Generated by `experiments/026-metabolism-flux/scripts/plot_molecule_mapping.py`.

Table 17: The eight species the composition check found priced as a different molecule. The accession route accepted the first identifier that resolved to a compound carrying a formation energy, whatever molecule that compound is; checking the resolved structure's composition against the model's own formula over all 2,414 accession-priced species finds 2,087 matches, 12 dolichols carrying exactly one extra isoprene unit, and these eight. Six are withdrawn; the glycogen pair is re-derived per residue from a curated polymer convention instead.

Species	Name	GEM formula	Priced as	$\Delta_f G'^{\circ}$	Outcome
`s_0773`	glycogen	`C6H10O5`	`C24H42O21`	-1127.2	re-derived
`s_0774`	glycogen	`C6H10O5`	`C24H42O21`	-1370.9	re-derived
`s_1578`	zymosterol intermediate 1c	`C28H43O3`	`C29H46O2`	1257.2	withdrawn
`s_2920`	3-oxo-cis-hexadec-9-enoyl-...	`C37H62N7O18P3S`	`C37H60N7O17P3S`	-575.1	withdrawn
`s_2923`	3-oxo-cis-octadec-9-enoyl-...	`C39H66N7O18P3S`	`C39H64N7O17P3S`	-395.1	withdrawn
`s_4207`	trans-4-hydroxy-L-proline	`C5H9NO3`	`C5H4N4O4`	-119.0	withdrawn
`s_4295`	dimethylallyl phosphate	`C5H9O4P`	`C8H8O`	-512.4	withdrawn
`s_4313`	ITP	`C10H11N4O14P3`	`C4H4O4`	-2531.7	withdrawn

Table 18: The 16 hand-curated records. *Admission* is the strongest claim the sources support, and only *measured* and *tabulated* reach the table a model reads: a provisional or absent record is loaded, validated and reported without being priced, which is why five of them price nothing. *Closed* counts reactions the record itself closed, with in parentheses the reactions closed because the species cancels across the reaction instead. Each record carries its sources with a sha256 per source in `experiments/026-metabolism-flux/results/curation/`.

Record	Kind	Admission	Decomp.	Spec.	Src	Closed	Value
chitobiosyl_pp_-dolichol	polymer	tabulated	yes	1	14	0	repeat unit C36H64N2O17P2, no linkage stated
cytochrome_c	carrier	measured	n/a	2	10	7	E′0 272.0 mV vs SHE, n=1, Saccharomyces cerevi...
ethylnitronate	structure	tabulated	yes	1	11	0	a curated structure, drawn as one microspecies
glycogen	polymer	tabulated	refused	2	2	0	repeat unit C6H10O5, curated dGf′0 s_0773 -244...
heme_a	structure	tabulated	refused	3	7	1	a curated structure, drawn as one microspecies
heme_o	structure	tabulated	refused	1	7	1	a curated structure, drawn as one microspecies
mannan	polymer	tabulated	refused	2	11	3	repeat unit C6H10O5, curated dGf′0 s_1107 -248...
nitrophenyl_-phosphate	structure	tabulated	mixed	2	7	0	per metabolite: s_3860=a structure, s_3859=non...
pectin	polymer	tabulated	refused	1	18	0	repeat unit C6H7O6, alpha-(1->4) glycosidic, t...
siroheme	structure	tabulated	refused	1	11	0	a curated structure, drawn as one microspecies
triphosphate	pKa ladder	measured	refused	2	10	0 (1)	pKa ladder [7.77, 5.4]
cytochrome_b5	carrier	provisional	n/a	2	10	0	E′0 5.5 mV vs SHE, n=1, Bos taurus
ferroheme_b	pKa ladder	provisional	refused	1	9	0	pKa ladder [4.75, 4.15]
gpi_intermediate	formula	provisional	n/a	1	17	0	formula -> C55H100N3O39P3R2
methionine_s_oxide	not found	none	refused	3	10	0	nothing found
porphyrin	not found	none	refused	2	9	0 (1)	nothing found

of supplied constants above pH 7. **The reference microspecies is therefore whatever protonation state the caller wrote the structure in, and that choice is worth 776 kJ mol^{-1}.** Pricing 7 written lipids from their neutral structures left a residual of -777.5 kJ mol^{-1} against the same compounds in the cache; submitting the pH 7 major microspecies instead leaves $+0.13$, range 0.07 to 0.15. The difference sits in two entries of the 163-long group vector, one for the phosphate and one for the amine, so it is the untransformed group-contribution term rather than the transform. **Submit the pH 7 major microspecies.** The error is silent otherwise: standard InChI normalizes mobile protons, so the two conventions share an InChIKey and a single local cache returns the compound the first call created, which is why the comparison needs two caches that cannot see each other.

The validation set is the cache's own constants, which is what makes it a test. No open-source predictor was trained to reproduce ChemAxon, so the cache is an independent reference rather than a leaked one. 2,400 compounds are drawn from the 694,324 the cache holds, stratified on constant count, functional group, mass and whether the compound is lipid-like, and the 2,100 outside the pinned seed-17 300 are what every number below is measured on. The pinned 300 are excluded because ensemble membership was fitted on them.

The rule is a tail bound at every compartment pH, not an average error. A method is scored by the *worst* per-pH 90th percentile of $|\Delta_f G'^{\circ}(\text{predicted}) - \Delta_f G'^{\circ}(\text{cache})|$ over the eight pHs the pipeline transforms at, subject to a failure rate of at most 0.02, breaking ties on the pH 7.2 median and then on cost. Three choices in that sentence are doing work. The p90 rather than the mean, because the fill-in is a tail risk and one wrongly signed reaction potential is what a second-law hinge acts on. The worst pH rather than the mean over pHs, because a metabolite is priced at its own compartment's pH and a good cytosol score would otherwise hide a bad vacuole one. And a failure-rate bound rather than a scored error, because a method that cannot answer leaves a gap rather than a cheap answer.

No single tool is admissible except the worst one. Uni-pKa on the simple enumeration template carries the smallest error of any single tool and returns no ladder for 11.5 percent of the compounds; MolGpKa fails on 5.7 percent, and 126 of its 2,400 empty ladders come back with no error at all, 93 of them lipid-like. Only pKasolver-lite clears the failure bound alone, at more than twice everyone else's error. **So the selected method is a cascade rather than a tool**: Uni-pKa's answer where it has one, MolGpKa's where it does not, pKasolver-lite's where neither does. Composition rather than averaging, because a median is undefined at a rank where the best member returned nothing.

The choice between the three-member and the two-member cascade is a tie the rule breaks arbitrarily,

Table 19: Every candidate pK_a method against the eQuilibrator compound cache's own ChemAxon constants, on the 2,100 held-out compounds. The rule selects the smallest *worst* per-pH p90 of $|\Delta\Delta_f G'^{\circ}|$ among methods whose failure rate is at most 0.02, breaking ties on the pH 7.2 median and then on cost. Failure rate is the fraction of compounds a method returns no ladder for, and it is a bound rather than a scored error because a method that cannot answer leaves a gap rather than a cheap answer. *Right count* is the fraction whose ladder length matches the cache's, reported and deliberately outside the rule: a ladder can carry the right number of constants and the wrong values, and the transform is what the flux layer consumes.

			worst pH		at pH 7.2			
Method	Kind	Fail	p90	pH	Median	Mean	Right count	Verdict
median of two	ensemble	0.1695	7.83	5	0.502	1.47	0.282	rejected, failure rate 0.170
cascade, three members	ensemble	0.0024	8.33	7.8	0.067	1.04	0.338	**selected**
cascade, two members	ensemble	0.0029	8.33	7.8	0.068	1.04	0.338	ranked below
Uni-pKa, simple template	tool	0.1152	8.75	7.8	0.077	1.16	0.354	rejected, failure rate 0.115
best tool per functional group	ensemble	0.1152	8.75	7.8	0.077	1.16	0.354	rejected, failure rate 0.115
MolGpKa	tool	0.0571	9.33	7.8	0.130	1.10	0.217	rejected, failure rate 0.057
median of three	ensemble	0.0048	9.38	7.8	0.331	1.80	0.364	ranked below
pKasolver-lite	tool	0.0071	18.12	7.8	1.466	4.05	0.364	ranked below

p90, median and mean are of $|\Delta\Delta_f G'^{\circ}|$ in kJ mol^{-1}.

Table 20: The p90 of $|\Delta\Delta_f G'^{\circ}|$, in kJ mol^{-1}, at each pH the pipeline transforms a metabolite at. A metabolite is priced at its own compartment's pH, so a method has to hold at all eight; scoring at pH 7.2 alone understates the selected cascade's tail by 3.4 kJ mol^{-1} and ranks MolGpKa above Uni-pKa. The mitochondrial matrix at pH 7.8 is the worst column for every method but one.

	compartment pH							
Method	5	6.2	6.6	7	7.1	7.2	7.4	7.8
median of two	7.83	2.19	2.17	3.50	4.01	4.48	5.46	7.76
cascade, three members	5.58	1.88	2.50	3.94	4.38	4.88	5.98	8.33
cascade, two members	5.58	1.89	2.50	3.95	4.39	4.89	5.98	8.33
Uni-pKa, simple template	5.56	1.94	2.75	4.39	4.89	5.41	6.48	8.75
best tool per functional group	5.56	1.94	2.75	4.39	4.89	5.41	6.48	8.75
MolGpKa	8.46	2.39	2.83	3.36	3.56	4.00	5.38	9.33
median of three	9.34	2.66	2.65	4.06	4.60	5.25	6.46	9.38
pKasolver-lite	12.58	3.87	3.88	7.02	8.28	9.63	12.27	18.12

8.331 against 8.333 kJ mol^{-1}, a difference of 0.02 percent. The rule picks the three-member cascade because it is deterministic; the two-member one is the defensible cheaper choice and nothing measured here separates them.

An earlier 300-compound benchmark at pH 7.2 alone ranks the tools the same way and carries three breakdowns the 2,400-compound run does not repeat, per functional group, per ladder length, and each tool's shift against each other tool's (Table 21, Figure 14). Two of its findings are load bearing. The enumeration template rather than the network decides Uni-pKa's answer: the same weights and the same code on the full template give 2.94 kJ mol^{-1} against the simple template's 0.91, because the enumerator walks outward from the neutral parent and the rungs it adds are near pH 7 by construction, which is where the transform is most sensitive. And the tools do not agree with each other: MolGpKa against pKasolver-lite gives $r = 0.085$, with a median absolute difference between two tools' shifts larger than the best tool's own median error against the cache.

Table 21: Open-source pK_a predictors against the compound cache's own ChemAxon-derived constants, over 300 sampled compounds at pH 7.2, I = 0.25 M, T = 303.15 K, pMg 3. The last three columns are the change in $\Delta_f G'^{\circ}$ when the predicted constants replace the cache's, with the reference microspecies and the untransformed formation energy held fixed, so they isolate the pK_a contribution. **The two Uni-pKa rows are the same weights and the same code**, and differ only in the microstate enumeration template, which fixes how many rungs a ladder can have before the network is consulted. **QupKake is unmeasured on this set**: a rung costs a full xTB wave, the run reached 4 of up to 14 rungs, and only 1 of 60 ladders had terminated. Its truncated row scores all 60 with the rungs that ran, so those numbers bound it rather than measure it.

						$\Delta_f G'^{\circ}$ error (kJ mol^{-1})		
Tool	n	Scored	Right count	pK_a MAE	MAE	Median	p90
pKasolver-lite	300	298	0.356	2.59	2.52	0.56	6.40
MolGpKa	300	294	0.248	2.83	1.21	0.34	3.20
Uni-pKa, full template	300	300	0.200	2.54	2.94	1.11	7.93
Uni-pKa, simple template	300	299	0.324	2.79	0.91	0.16	2.84
QupKake	60	1	0.000	4.56	1.87	1.87	1.87
QupKake, truncated	60	60	0.283	3.61	1.35	0.38	5.08

A ladder that is wrong is worth about ten kilojoules, and a reference microspecies that is wrong is worth about eight hundred. Those two are the whole error budget of this route, and they are not comparable in

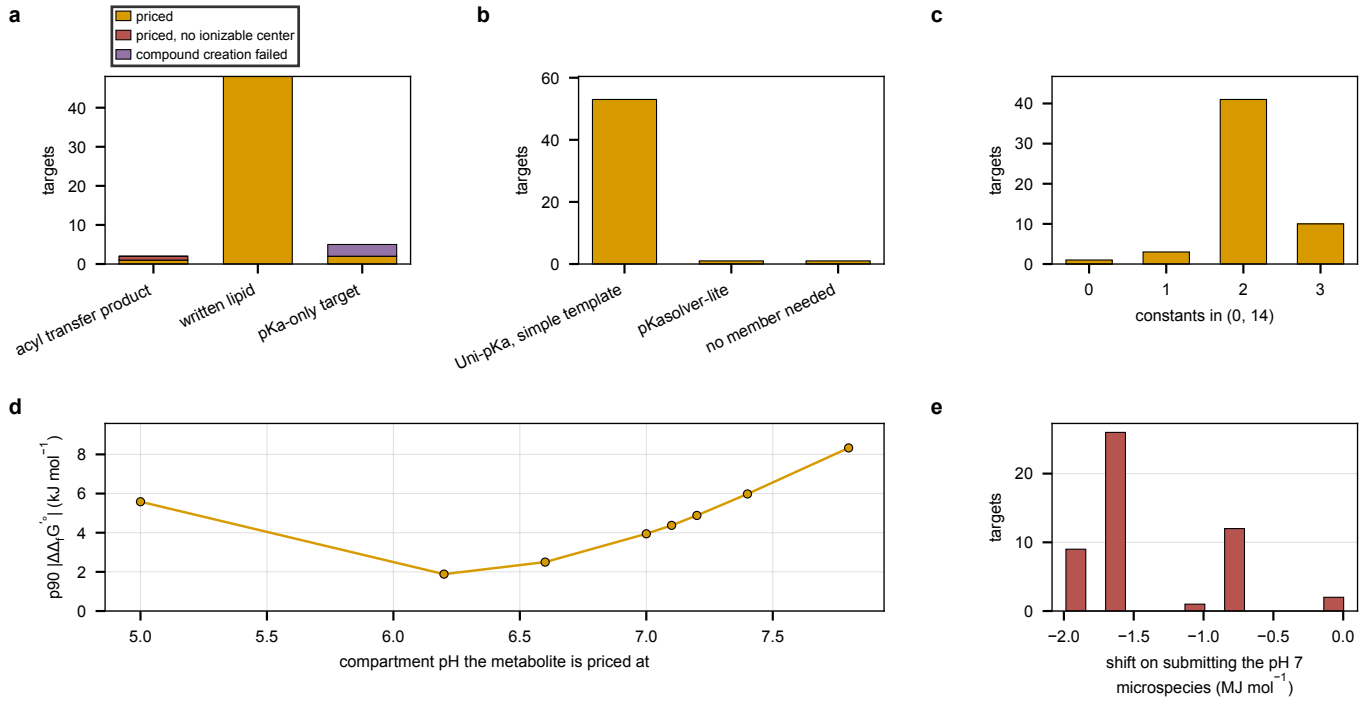


Figure 13: What the selected cascade was asked to do on yeast-GEM 9.0.2, and how large the two shifts a ladder controls are. A target is one distinct structure; several model species can share one, and the 52 targets that yield a value carry the 70 metabolite rows the route table counts, 68 of them from a target with an ionizable center. **a**, The 55 targets by where they came from, partitioned by outcome. 48 are lipids the structure writer composed and the cache does not hold, 5 are species needing only a ladder, and 2 are acyl-transfer products. Three fail at compound creation and one is created as a single microspecies because a SMARTS scan finds no ionizable center in it, which is the correct answer rather than an error. **b**, Which cascade member answered. Uni-pKa on the simple template served 53 and pKasolver-lite 1, so the cascade degenerates to its first member on this population and earns its place on the benchmark rather than here. **c**, Ladder length, the number of constants inside pH 0 to 14 the cascade returned. Two is the mode, as expected for a phosphodiester lipid headgroup. **d**, The Legendre-transform error a ladder carries, as the p90 of $|\Delta\Delta_f G'^{\circ}|$ at each of the eight working compartment pHs over the 2,100 held-out benchmark compounds. The mitochondrial matrix at pH 7.8 is the worst column, 8.33 kJ mol^{-1}, against 4.88 at pH 7.2, which is why the rule reads the worst pH rather than the cytosolic one. **e**, The reference-microspecies shift, which is three orders of magnitude larger than the ladder error: the signed change in each target's transformed formation energy when the pH 7 major microspecies replaces the neutral parent the writer emitted, median −1,585 kJ mol^{-1} over the 50 targets priced both ways. Generated by `experiments/026-metabolism-flux/scripts/plot_pka_outside_equilibrator.py`.

size. The ladder error is measured against the cache's own constants and runs to a p90 of 8.33 kJ mol^{-1} at the worst compartment pH; the reference-microspecies error is a fixed offset in the group-contribution term and moved the 48 written lipids by a median of 1,585 kJ mol^{-1} when it was corrected. A worked case of the first is triphosphate: the cache stores a six-microspecies ladder one rung too far in both directions, and against the measured ladder the two Legendre transforms differ by **12.762 kJ mol^{-1}**. That is what the cached ladder would inject into `r_4403` the day an anchor for triphosphate exists, so the measured ladder is recorded now and prices nothing on its own.

Magnesium is the gap this route cannot close. `specified_pkas` carries proton dissociation constants only, and `Compound.transform` reads magnesium through the microspecies table that is materialized once at creation, so a constant appended afterwards commits and changes nothing. Omitting them costs 0.0018 to 4.12 kJ mol^{-1} at pMg 3 on the benchmark compounds that carry any, always in the direction of under-stabilizing the compound, and up to 9.43 at pMg 2. Carrying a constant over by substructure is worse than omitting it: bucketing eQuilibrator's own 535-compound table by ionizable phosphate steps leaves a within-bucket standard deviation of 1.6 to 3.3 log units, and a pK_d off by 2 costs 10.87 kJ mol^{-1} on 5-formyl-THF and 11.14 on ADP against 4.12 and 1.10 for having no constant at all.

One interface defect belongs with these. The guard tests truthiness rather than presence, so an empty list, the correct answer for a compound with no constant between pH 0 and 14, raises rather than creating a single-microspecies compound. Downstream the route tests for an ionizable center by SMARTS instead of reading a tool's error string, because the three cascade members word "nothing here" three different ways and one of them words it as an error.

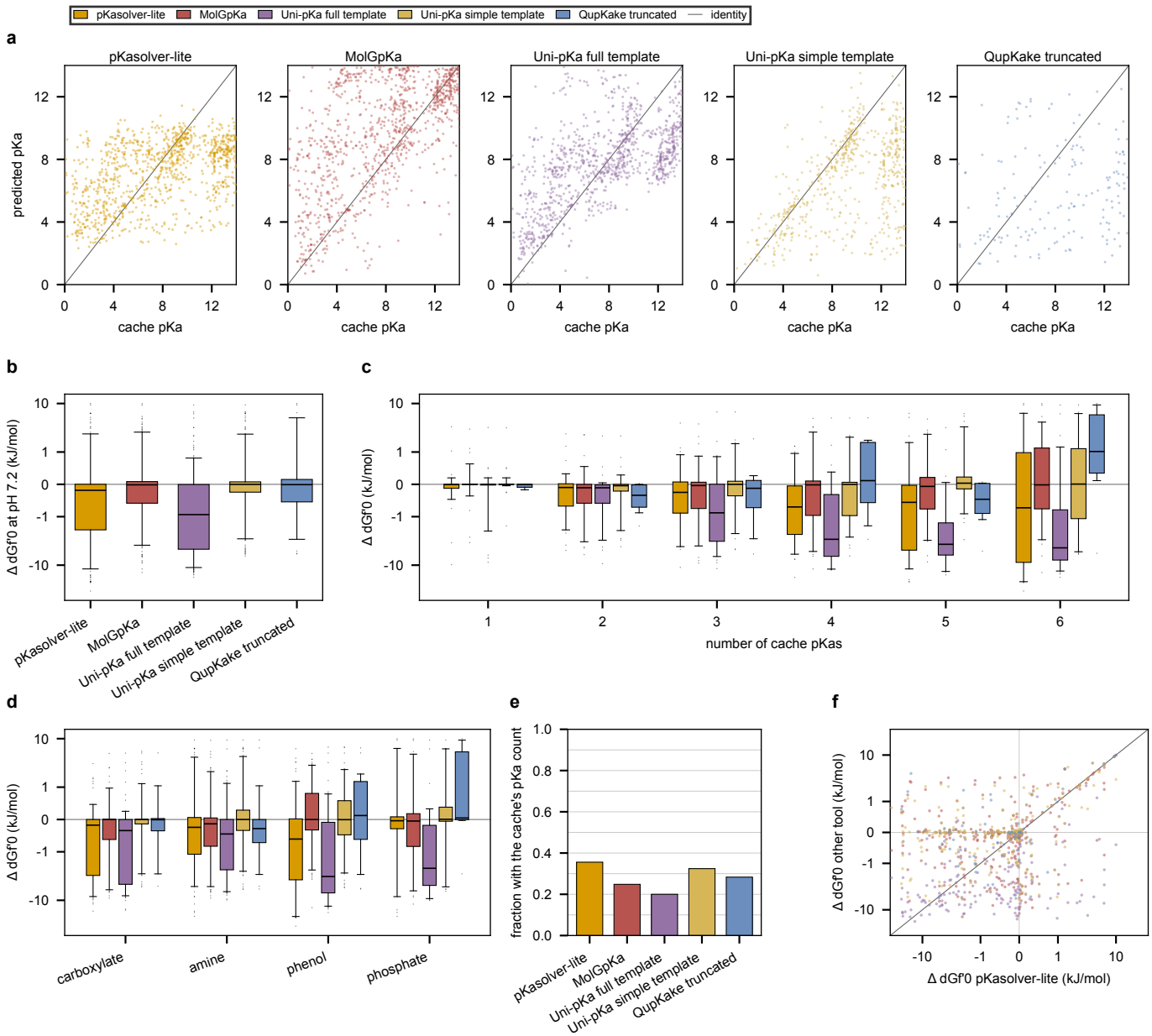


Figure 14: Four open-source pK_a predictors against the compound cache's ChemAxon constants, on 300 compounds at pH 7.2. This is the earlier and smaller benchmark; the selection of Table 19 runs on 2,100 held-out compounds at eight pHs and supersedes its ranking. **a**, Predicted against cache pK_a, paired by rank over the shorter of the two lists, on 1,010 pairs for pKasolver-lite, 1,021 for MolGpKa, 1,043 for the full template, 596 for the simple template and 167 for truncated QupKake; the simple template is the only cloud that follows the identity line, with 0.374 of its pairs inside 1 pK_a unit against MolGpKa's 0.358, the full template's 0.267 and pKasolver-lite's 0.258, on half as many pairs. **b**, Signed change in $\Delta_f G'^{\circ}$ per compound; every median sits just below zero, so what separates the tools is tail width, at 5th percentiles of -12.19 kJ mol^{-1} for pKasolver-lite, -11.22 for the full template, -3.93 for MolGpKa and -2.87 for the simple template. **c**, The same shift split by the number of constants the cache holds, where the full template's error grows monotonically from 0.64 to 5.43 kJ mol^{-1} across the six strata while the simple template's spans 0.14 to 2.13. **d**, The same shift split by functional group; the simple template is the smallest in all four, 1.03 against MolGpKa's 1.35 on amine, 0.71 against 1.00 on carboxylate, 1.08 against 1.18 on phenol and 1.75 against 2.33 on phosphate. **e**, Fraction of compounds whose constant count matches the cache's, 0.356 pKasolver-lite, 0.324 simple template, 0.248 MolGpKa, 0.200 full template, which ranks the tools differently from the energy columns. **f**, Each tool's per-compound shift against pKasolver-lite's, which is the question of whether the predictors agree with one another: they do not, at $r = 0.085$ for MolGpKa and 0.268 for the simple template, with median absolute differences of 0.74 and 0.62 kJ mol^{-1}, larger than the simple template's own median error against the cache. QupKake is drawn only as its truncated row. Generated by `experiments/026-metabolism-flux/scripts/plot_external_pka.py`.

5.5 Writing the missing lipid structures ✗

The 409 metabolite rows with a definite formula and no structure in any namespace are lipids whose acyl composition is written into the name, so the structure is derivable from the name even though no database indexes it. `lipid_structure_writer.py` parses each name into a headgroup template and a list of acyl chains, assembles the SMILES in RDKit, and prices the result two ways.

168 of the 206 distinct names parse, covering 347 of the 409 metabolite rows, and all 168 reproduce yeast-GEM's own elemental formula exactly. The 38 that do not split into 17 SLIME backbone pseudo-metabolites, whose formula is a difference formula rather than a molecule, and 21 names with no acyl composition to read: glycans, hemes, cytochromes, protein-linked lysines and a GPI anchor. Every double bond is placed at Δ-9 and cis, a convention yeast-GEM's `16:1` does not state; the written free acids hash to the InChIKeys of palmitoleic and oleic acid, which is what the build asserts rather than the convention.

All 168 price, 120 by finding a compound eQuilibrator already holds and 48 through the external pK_a route. The cache lookup is a route the coverage audit had no structure to key on, and it checks the structures independently: 120 written InChIKeys resolve a compound nobody here drew.

What this buys is method consistency rather than coverage. The eQuilibrator route moves from 66.35 to 86.27 percent of reactions, passing the shipped column's 84.85, and every one of those values now comes from one group-contribution method with an uncertainty attached. **The union with the shipped table does not move at all**, staying at 3,716 reactions, because that table already prices these lipids by name out of yETFL and ModelSEED.

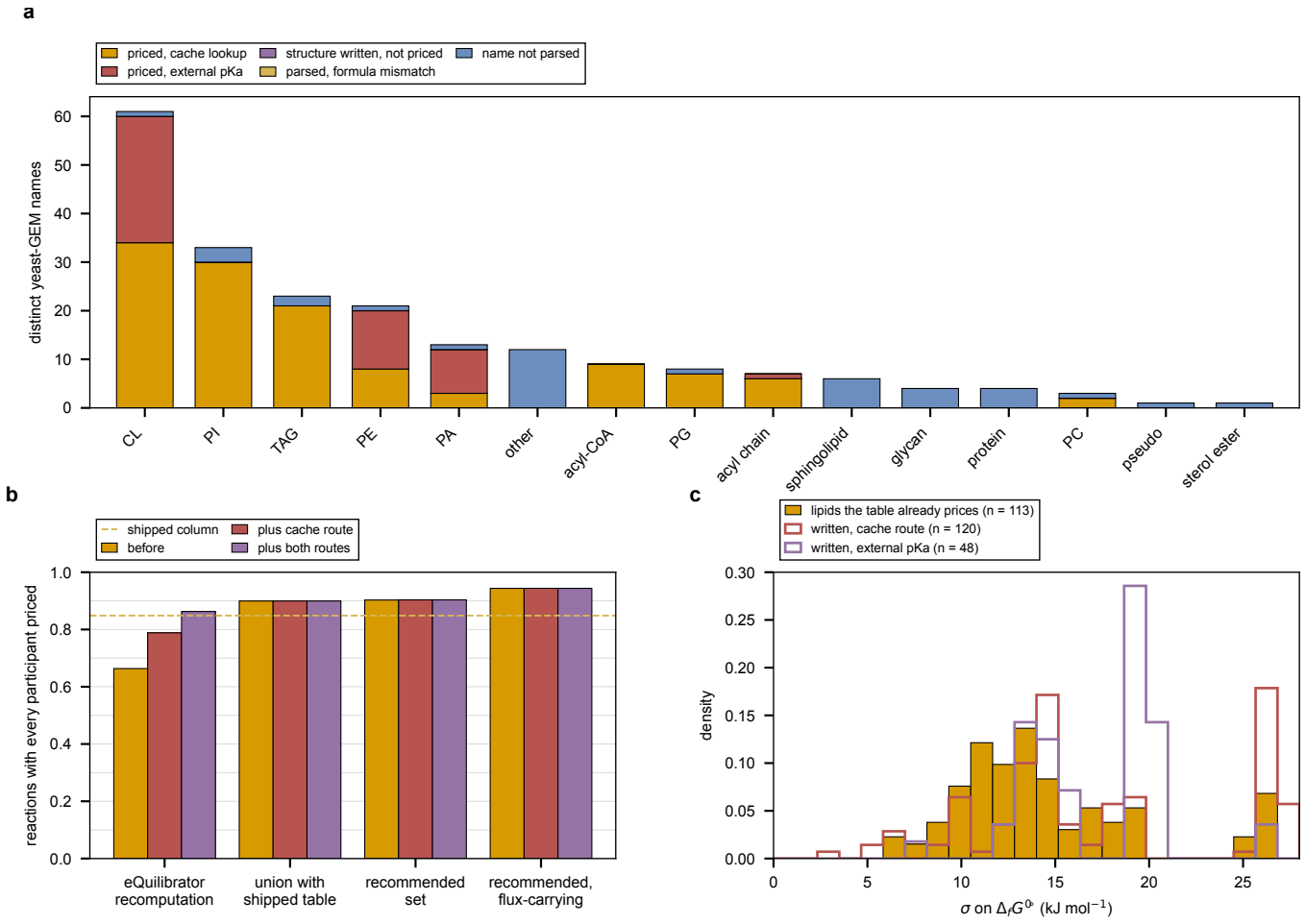


Figure 15: What the lipid structure writer reached, and what it bought. a, The 206 distinct names by yeast-GEM chemical class, each bar partitioned into the five mutually exclusive stages of the pipeline. No name is lost to a formula mismatch and none to a written structure that failed to price, so those two segments are empty everywhere; the whole loss is the 38 names the parser refuses. **b**, Reactions of 4,131 with every participant priced, at four levels of aggregation, before the written lipids, with the 120 names that resolve an existing cache compound, and with all 168. Dashed, the shipped metabolite column's 84.85 percent. The eQuilibrator recomputation crosses it; the union with the shipped table and the flux-carrying restriction do not move. **c**, Uncertainty on the formation energy, one value per distinct name: the 113 lipid names the recomputed table already prices, the 120 written names priced by cache lookup, and the 48 priced through a created compound. Medians 13.19, 14.79 and 18.92 kJ mol^{-1}, so a written structure is priced no less certainly than a lipid the table already held. Generated by `experiments/026-metabolism-flux/scripts/plot_lipid_structures.py`.

5.6 Composition matching cannot buy coverage ✗

455 unpriced metabolites have a heavy-atom composition that the compound cache can already decompose, which invites pricing them from a same-composition surrogate. The test is to price every cache compound of the same composition and measure how far apart they land.

They land far apart. Across the 441 metabolites that gain a value, the median spread within a composition is **52.6 kJ/mol** and the 90th percentile is 342 kJ/mol. Only **39** fall below 5 kJ/mol. At 303 K, $RT \ln 10$ is 5.8 kJ/mol, so the median surrogate is uncertain by roughly nine orders of magnitude in the equilibrium constant, which is not a constraint on a sign. Wiring all 441 in would report coverage of 93.49 percent, three and a half points above the union, on values that cannot orient a reaction. The 39 tight ones are offered and 12 are taken by the built table, the rest refused: **a resolution rate and a usable rate are different quantities**, and only the second constrains a model. The 5 kJ/mol admission bound is a judgment rather than a measured threshold, $RT \ln 10$ rounded down.

5.7 What is left, and which of it is permanent ✗

Table 22: Every class that holds a value back, the species it holds and the reactions it blocks. *Species* counts metabolites carrying that reason; *blocks* counts reactions with at least one such participant, so the column double counts a reaction blocked by two classes; *sole* counts reactions this class alone blocks and partitions the 330 unpriced reactions exactly. A class marked permanent is one no pKa predictor, structure writer, decomposer or literature table can move, so the sum of the permanent rows is the ceiling for any method rather than for this one.

Class	What it means	What it would take	Species	Blocks	Sole
pseudo-metabolite†	a modeling device, a pool, a SLIME backbone or a difference formula, rather than a molecule	none; the ceiling	55	213	201
placeholder formula†	an R, X or ∗ in the formula: the species is a family of molecules and no single formation energy exists for it	none; the ceiling	159	99	89
decomposition refused	the structure exists and component contribution refuses it, so a pKa predictor changes nothing	a decomposition, or a tabulated value	16	13	11
no structure	a definite formula with no structure in any namespace and no writer that composes one from the name	a structure from a namespace or a curator	6	10	10
more than one reason	the reaction is blocked by species in two or more of the classes above	every blocking class resolved	0	0	14
placeholder structure†	the only structure that resolves carries an attachment point, so the species is a residue inside a macromolecule	none; the ceiling	6	3	2
recorded, not admitted	a curation record exists and its admission is computed or provisional, so it is reported and not priced	a measured or tabulated source	4	5	2
group in the null space	the structure decomposes and one group is in the component-contribution null space, so the estimate has infinite variance	a measurement of a compound carrying that group	1	1	1
non-chemical reaction†	an exchange, demand, sink or biomass pseudoreaction, not elementally balanced, so a standard reaction energy has no meaning for it	none; excluded from the chemical denominator	0	0	0

† permanent under the admission rule.

291 of the 330 unpriced reactions are closed for good. A reaction is permanently closed when every metabolite blocking it is not a molecule: a placeholder formula naming a family, a SLIME backbone whose formula is a difference formula, or a residue inside a macromolecule. No pK_a predictor, structure writer, decomposer or literature table changes any of those. The 21 that remain are what the last two rungs of Figure 11e would close, and the probe that measured them found none of the refused structures reachable by the reactant-contribution path or by a surrogate inside the admitted spread, so that rung needs new data rather than new code.

Five chemistries are permanently refused for a reason that is about eQuilibrator's group system rather than about the species. The porphyrin macrocycle has no group, so heme a, heme o, siroheme and ferroheme b decompose nowhere and are closable only at the reaction level where the macrocycle cancels. The sulfoxide of L-methionine S-oxide has no group. 4-nitrophenyl phosphate, an aryl phosphate, has none either. The nitro group is the opposite case and is worth separating: the group *exists* and sits in the component-contribution null space, carried by 0 of 631 training compounds, so the estimate has infinite variance rather than no value. Triphosphate is a fourth kind again: it decomposes nowhere and appears in none of the 3,999 training reactions, so the reactant-contribution path that prices phosphate and diphosphate despite their empty group vectors is closed to it too. Placeholder and R-group species are the fifth, and are a property of the model.

Five defects in yeast-GEM 9.0.2 were found while curating, and they are upstream rather than ours. Heme a's declared charge of −3 is chemically unreachable: no microspecies of `C49H55FeN4O6` carries it, the ferrous dianion being `C49H54FeN4O6` at −2, so every curated structure fails a strict charge comparison against the model. `r_0530`'s stoichiometry disagrees with EC 1.17.99.9, which matters beside the −440.539 kJ mol^{-1} the conserved-moiety route prices it at. Pectin `s_1309` is written at charge 0 and the galacturonate residue is −1, which is why that reaction fails a charge balance. `s_3934`, named "Ethylnitronate", is nitroethane: the reaction it appears in balances only with `C2H5NO2` at charge 0, and the nitronate anion leaves it short one hydrogen and one negative charge. And `s_0332`'s GPI formula was invented by a solver in 2018 rather than derived, which is why walking the pathway from the last intermediate that carries a formula gives a different one. The dolichol convention above is a sixth, and unlike these five it is internally consistent and merely undocumented.

5.8 The kinetic tables, tier by tier ✗

Table 23: The four tiers a turnover number or Michaelis constant is drawn from, tried in this order over the 3,728 catalytic units of yeast-GEM 9.0.2. k_{cat} and K_M resolve on the same keys and land on the same counts, because a unit that has one measured has the other. The homolog tier is measured and is nonetheless folded in with the predictions wherever an experimental fraction is reported, since a measurement of another organism's enzyme is not a measurement of this one.

Tier	Keyed on	Evidence about	Units	Share
measured, this organism	the unit's own UniProt accession in the Open Enzyme Database mirror, and the assay substrate matched by InChIKey connectivity block to a substrate the reaction consumes	this protein doing this chemistry	97	2.6%
measured, same EC, another organism	an EC number annotated on the reaction plus the same substrate match, in any organism	the chemistry, not this protein, so it is never counted as a measurement of the modeled organism	782	21.0%
predicted	the protein sequence and the substrate structure	the median over the predictors the within-class study admits, which is EZKinetics, EITLEM and UniKP for k_{cat} and EZKinetics alone for K_M	2,831	75.9%
organism default	nothing; it is the fallback	the median of the values the database tiers placed in this same table, so it is a summary of the organism rather than of the unit	18	0.5%

Two and a half percent of the model's turnover numbers are measurements of its own enzymes, and a further 21 percent are measurements of the same chemistry somewhere else. The homolog tier is the largest single change of this round and it is deliberately not counted as experimental: a cross-organism EC match is a real measurement of a real enzyme, and not of the enzyme in this model. Where both are available it is also the better estimate, closer to the measured value than the predictor median on 55 of the 87 units carrying both, at a median absolute error of 0.094 decades against 0.300.

Eighteen units reach the organism default and none of them is reachable with an open-source tool. Fourteen of the 28 substrate rows behind them carry an R-group placeholder, which is a curation decision in the model rather than a missing structure; seven are macromolecular species with no formula, which need a predictor that accepts a protein substrate and all seven mirrored models take a small-molecule SMILES; five have no candidate structure in any mirrored catalog and the substrate writer was run on them and refused each one; two are cytochrome c, where the model writes only the heme part of a protein-bound species. **A substrate the writer tried and refused counts as unreachable whatever its reason class**, which is what makes this a ceiling rather than a backlog.

5.9 The kinetic ceiling is substrate structure, not the models ✗

DLKcat, UniKP, EITLEM and EZKinetics cover exactly the same 3,552 of 3,728 catalytic units on the unverified tables, and DeepEnzyme covers 3,546. Four independently trained models agreeing to the unit is not a coincidence about the models: every gene has a sequence, so the binding constraint is whether the substrate has a structure to featurize. **The 95.3 percent ceiling is a property of the inputs, and no better predictor moves it.** Two structure routes moved it instead. Resolving an outdated MetaNetX cross-reference priced 27 substrates, and supplying the lipid writer's own structures priced 48 more, between them closing 165 of the 176 units that had no prediction at all. TurNuP's 55.2 percent is a model limitation rather than an input one (Section 4.6).

The consensus tables take the per-unit median across the admitted predictors in log space and carry the across-predictor spread beside it. **For k_{cat} that spread has a median of 0.74 orders of magnitude over the three admitted predictors and the 3,698 units all three price**, down from 1.14 across all six, and 29.9 percent of units still

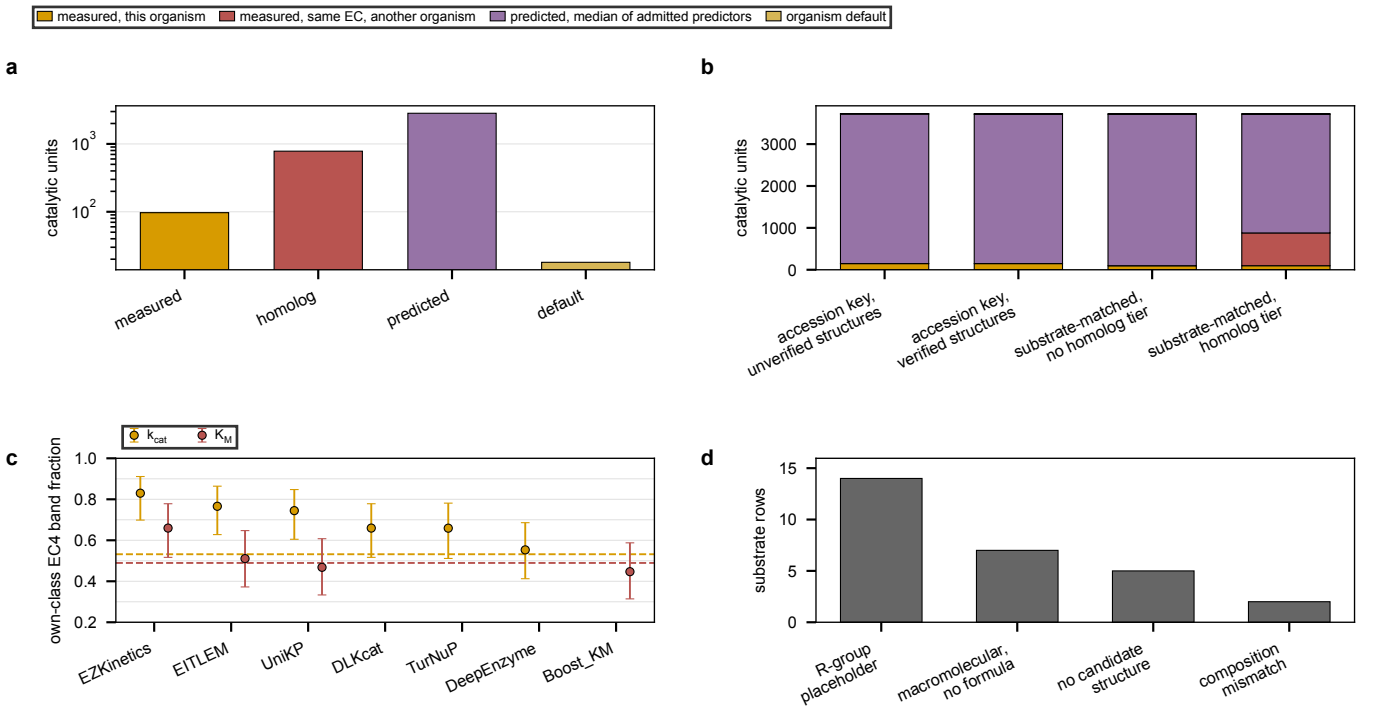


Figure 16: What the resolved k_{cat} and K_M tables are made of, and the rule that decides which predictors may fill them. **a**, The four tiers over the 3,728 catalytic units, on a logarithmic count axis. 97 units carry a measurement of the modeled organism's own protein on a substrate the reaction consumes, 782 a measurement of the same chemistry in another organism, 2,831 a predictor median and 18 the organism default. **b**, The same counts under each table version. Keying the measured tier on a substrate match rather than on the accession alone takes it from 148 units to 97, which is a correction rather than a loss: 51 of the 148 accession matches were assaying a substrate the reaction does not consume. Verifying every substrate structure moved 8 units onto the default. The homolog tier recovers 33 of those 51, whose chemistry is measured elsewhere, and 18 fall to a predictor. **c**, The admission rule. Each point is a predictor's fraction of measured pairs landing inside the band of enzymes sharing its own EC4 class, with the study's Wilson 95 percent interval; the dashed line is the EC-median lookup's point fraction for that parameter. A predictor is admitted only when its lower bound clears the line, which keeps EZKinetics, EITLEM and UniKP for k_{cat} and EZKinetics alone for K_M. EITLEM's K_M lower bound is 0.372 against the lookup's 0.489, so it is refused on the parameter it is admitted on for turnover. **d**, The 28 substrate rows behind the 18 units still on the default, by why no open-source tool reaches them. Generated by `experiments/026-metabolism-flux/scripts/plot_kinetics_tiers.py`.

disagree by more than a factor of ten. It is concentrated rather than uniform: units whose substrate has fewer than 12 heavy atoms sit at 0.55 decades and those above 60 heavy atoms at 1.38, and two-gene units at 1.20 against 0.71 for single-gene ones. A k_{cat} consensus at 0.74 is still a factor of five, so the table is an order-of-magnitude prior rather than a measurement. For K_M the admitted set is one predictor, so there is no spread to report and the honest uncertainty is the one the within-class study measures rather than an agreement statistic.

5.10 Changelog ✗

2026-09-11. Coverage closed. Formation energies: 3,801 of 4,131 reactions priced, 3,535 of the 3,847 chemical ones and 2,724 of the 2,845 that can carry flux, each value carrying one of nine routes and each gap one of seven failure classes. Eight accession prices were found to be for a different molecule, six of them withdrawn. Sixteen curation records landed, eleven admitted. Kinetics: 97 measured, 782 homolog, 2,831 predicted and 18 default over 3,728 catalytic units, on tables whose every substrate structure passes a composition check. This section was rewritten around the route attribution; the earlier version reported the recomputation at 66.4 percent, which is now the baseline row of Table 13 rather than the result.

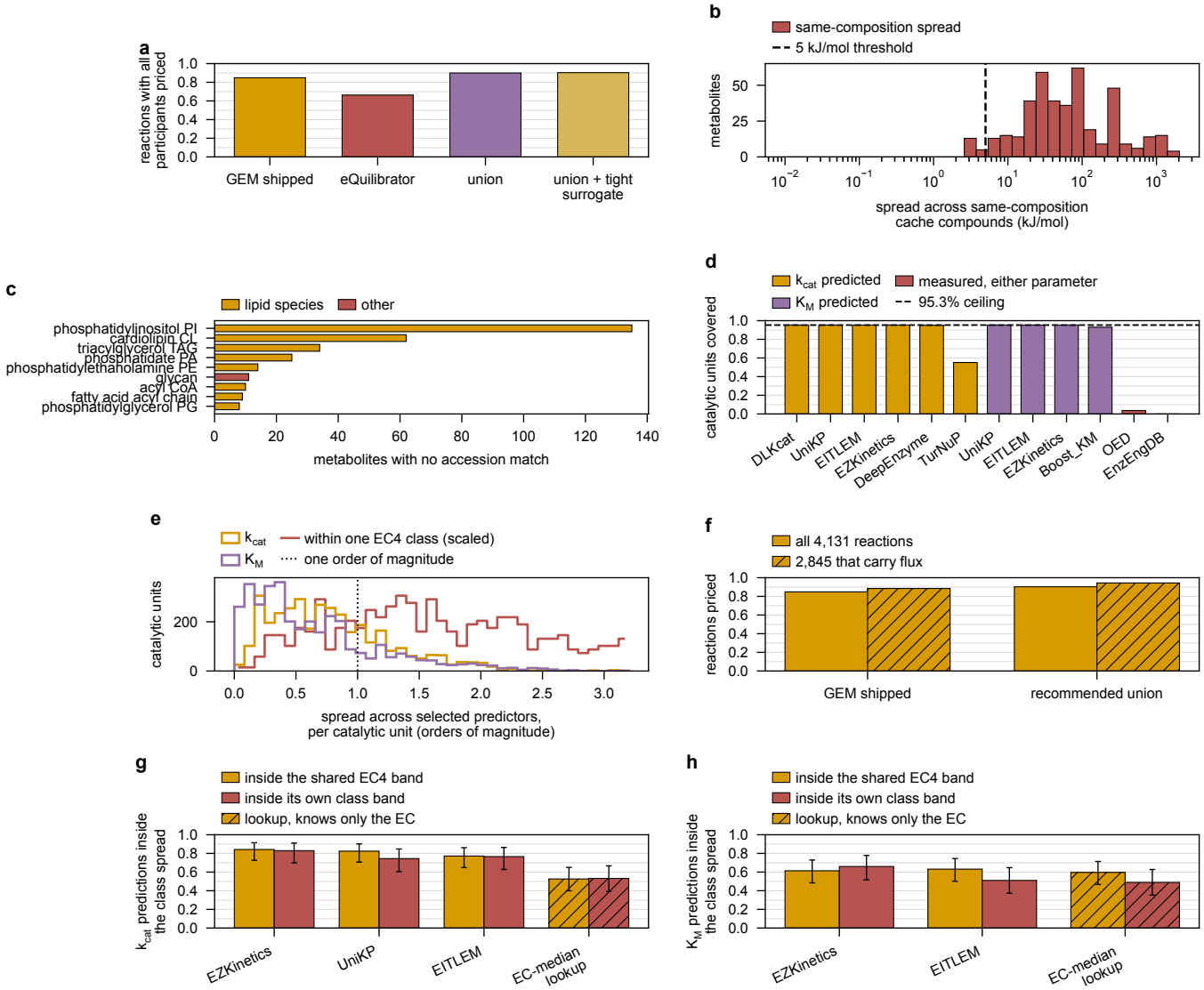


Figure 17: Coverage of the three parameter tables across yeast-GEM 9.0.2, and what coverage means. a, Reactions with every participant priced, by source. The union of the shipped table and the eQuilibrator recomputation exceeds either alone by 211 reactions, because the two price different reactions rather than nested ones. **b**, Spread of formation energy across cache compounds sharing a metabolite's heavy-atom composition. The dashed line is the 5 kJ/mol threshold below which a composition is taken to determine the energy; 39 of 441 clear it, so composition matching is rejected as a coverage route. **c**, Metabolites with no accession match, by chemical class; orange is a lipid species and red is anything else. The gap is lipids: phosphatidylinositol, cardiolipin and triacylglycerol species alone account for 231 of 332. **d**, Per-predictor coverage of the 3,728 catalytic units. Five of the six k_{cat} predictors and three of the four K_M predictors sit at the same 95.3 percent ceiling, which is set by substrate structure availability rather than by any model. TurNuP's 55.2 percent is its own limitation. The two red bars are what measured sources supply on the accession key alone, before the substrate match of Section 5.8 corrects them: the Open Enzyme Database prices 138 units, 3.7 percent, and EnzEngDB prices none, since it holds no k_{cat} or K_M field and no *S. cerevisiae* content (Section 4.6). **e**, Across-predictor spread per catalytic unit, over the *admitted* predictors of panels g and h. Restricting to three k_{cat} predictors takes the median spread from 1.14 to **0.72 orders of magnitude** and the fraction of units disagreeing by more than a factor of ten from 61.8 to 29.2 percent. The red outline is the spread across the distinct enzymes of one EC4 class, rescaled since it counts 523 classes rather than 3,698 units, so its shape and not its height is the comparison. **Even after selection the k_{cat} spread sits inside the biological one rather than below it. f**, The same coverage counted over all reactions and over only the 2,845 that can carry flux at nonzero FVA width. Restricting to reactions the medium permits raises the recommended union from 90.3 to 94.4 percent. **g**, Within-class plausibility for the k_{cat} predictors, against the EC-median lookup drawn hatched. The rule reads the Wilson lower bound against the lookup's point fraction of 0.532: EZKinetics 0.699, EITLEM 0.628 and UniKP 0.605 clear it, DLKcat 0.517, TurNuP 0.511 and DeepEnzyme 0.412 do not. Accuracy on the 57 measured pairs is deliberately not in the rule, since a recalled value lands inside a class band no more readily than a correct one. **h**, The same for K_M, where the rule keeps **one**: EZKinetics clears the lookup's 0.489 at a lower bound of 0.517, and EITLEM 0.372, UniKP 0.333 and Boost_KM 0.314 do not. A predictor a table lookup beats is not evidence about the enzyme.

Table 24: Every catalytic unit still carrying the organism default, and the substrate that keeps it there. A unit needs a structure for at least one substrate its reaction consumes; these have none any mirrored predictor can take. *Rows* is how many of the unit's substrates fail. None of the four classes is reachable with an open-source tool: an R-group placeholder is a curation decision in the model rather than a missing structure, a macromolecular substrate needs a predictor that accepts a protein and all seven mirrored models take a small-molecule SMILES, and the structure writer was run on the last two classes and refused them.

Unit	Reaction	Genes	Substrate	Why	Rows
556	r_0520	YKL165C	6-O-[alpha-D-mannosyl-(1->4)-alpha-D-g; phosphatidylethanolamine backbone	macromolecular species with no formula in the GEM	2
559	r_0523	YDR302W	6-O-{alpha-D-mannosyl-(1->2)-alpha-D-m; phosphatidylethanolamine backbone	macromolecular species with no formula in the GEM	2
560	r_0523	YLL031C	6-O-{alpha-D-mannosyl-(1->2)-alpha-D-m; phosphatidylethanolamine backbone	macromolecular species with no formula in the GEM	2
1379	r_1228	YBR180W	N,N'-diformyldityrosine	no candidate structure in any mirrored catalog for this ...	1
1565	r_2148	YHR067W	3-hydroxybutanoyl-ACP	R-group placeholder, the GEM defines the species only up...	1
1566	r_2149	YHR067W	3-hydroxyhexanoyl-ACP	R-group placeholder, the GEM defines the species only up...	1
1567	r_2150	YHR067W	3-hydroxyoctanoyl-ACP	R-group placeholder, the GEM defines the species only up...	1
3277	r_4152	YAL039C	Cytochrome c	candidate structures exist but none matches the GEM's co...	1
3278	r_4152	YKL087C	Cytochrome c	candidate structures exist but none matches the GEM's co...	1
3414	r_4239	YGL017W	Arg-tRNA(Arg); generic protein	R-group placeholder, the GEM defines the species only up...	2
3458	r_4277	YLR099C	1-acyl-sn-glycerol 3-phosphate; Acyl-CoA	R-group placeholder, the GEM defines the species only up...	2
3459	r_4277	YPR139C	1-acyl-sn-glycerol 3-phosphate; Acyl-CoA	R-group placeholder, the GEM defines the species only up...	2
3462	r_4280	YLR239C	Apoprotein; octanoyl-ACP	R-group placeholder, the GEM defines the species only up...	2
3463	r_4281	YLR239C	Apoprotein; Lipoyl-[acp]	R-group placeholder, the GEM defines the species only up...	2
3505	r_4325	YOR226C	S-sulfanyl-[L-cysteine desulfurase]; [disordered-form [Fe-S] cluster scaffo	macromolecular species with no formula in the GEM	2
3506	r_4325	YPL135W	S-sulfanyl-[L-cysteine desulfurase]; [disordered-form [Fe-S] cluster scaffo	macromolecular species with no formula in the GEM	2
3662	r_4700	YIL166C	hydrogen sulfide	no candidate structure in any mirrored catalog for this ...	1
3665	r_4704	YIL166C	alkanesulfonate	R-group placeholder, the GEM defines the species only up...	1

6 Where these tables enter the model ✗

Figure 18 places the flux module inside the cell graph transformer. The module is neither of the two instrument classes the architecture previously had: a Type I instrument maps a representation to a representation and a Type II instrument maps a representation to a measured output, while the flux module maps a representation to a *flux vector*, a latent physical state from which phenotypes are then read.

Three of its inputs are parameter tables rather than learned weights. The k_{cat} table bounds the box the flux is parameterized inside; the formation energies set the reaction potential whose sign the second-law penalty acts on; the K_M table is built and not yet enabled, for the reason Section 7 gives.

6.1 Why these two tables, in two equations ✗

k_{cat} is what lets a genotype change a flux. The genotype reaches the flux through a predicted enzyme level, not through a gate. Write h_g for the perturbed cell's embedding of GEM gene g, $\gamma_g \in \{0, 1\}$ for whether that gene is present, E_g^0 for a prior abundance in mmol gDW^{-1}, w for a learned readout vector and r for the range, in natural-log units, the head may move a gene away from its prior:

$$E_g \;=\; \gamma_g \, E_g^0 \, \exp\!\big(r \tanh(w^\top h_g)\big), \qquad r = 3. \tag{2}$$

The tanh bounds the deviation, so a gene stays within a factor of e^3, about 20, of its prior in either direction and no gene can be driven to zero by the head alone. A deletion is the one thing that can: it sets $\gamma_g = 0$ exactly. The

prior is equal-molar, $E_g^0 = \frac{1}{2} P_{\text{avail}} / \sum_g MW_g$, because no measured proteome is passed in; MW_g is the gene product's molecular weight in kDa and P_{avail} the protein pool defined below.

A catalytic unit is a complex, and a complex is limited by its scarcest subunit. Over the subunit set C_u of unit u, with $\beta = 8$:

$$E_u \;=\; \Big(\sum_{g \in C_u} E_g^{-\beta} \Big)^{-1/\beta}, \tag{3}$$

a power mean of exponent $-\beta$, which approaches $\min_{g \in C_u} E_g$ as β grows. **Deleting one subunit therefore takes the whole complex down, not a share of it**, which is what a heteromeric enzyme does and what a sum over subunits would get wrong.

The capacity bound is then the product of a turnover number and an amount of enzyme, summed over the units that catalyze the reaction. With $\rho(u)$ the reaction unit u serves:

$$|\nu_j| \;\le\; \sum_{u \,:\, \rho(u) = j} k_{cat,u}\, E_u. \tag{4}$$

Isozymes add, since either enzyme can carry the flux on its own. A reaction with no gene annotation is left uncapped, because an unannotated reaction is one nobody assigned a gene to rather than one no gene catalyzes.

The pool is finite, so the abundances are priced against a budget:

$$\sum_g MW_g E_g \;\le\; P_{\text{avail}}, \qquad C_{\text{bud}} \;=\; \Big[\sum_g MW_g E_g / P_{\text{avail}} - 1 \Big]_+, \qquad P_{\text{avail}} = 0.35 \text{ g gDW}^{-1}, \tag{5}$$

which is GECKO's constraint [9]. It is a *penalty* rather than a projection: nothing renormalizes E_g, so a trained model can overspend the pool and pay for it. That makes C_{bud} a reported quantity as well as a loss term, and it is what couples the genes to each other. A deletion frees mass that the remaining genes can take.

Only the product $k_{cat,u}E_u$ reaches the bound, and that fixes what the head can be said to predict. Equation (4) never sees E_u alone, so the head is identifiable only up to that product: a k_{cat} that is low by a factor of ten and an abundance that is high by ten give the same ceiling and the same loss. Read that way the head is predicting an enzyme *capacity*, and the k_{cat} table sets the units it is expressed in rather than pinning its value. It is also why the across-predictor spread of Section 4.6 is not simply absorbed: it moves $k_{cat,u}$ per unit, so it changes which reactions are tight relative to each other, and that is a shape the head has to undo gene by gene rather than with one scale factor.

K_M appears nowhere in Equations (2) to (5), and that is the code rather than a simplification for the page. The capacity is $k_{cat,u}E_u$ with nothing multiplying it, and a saturation factor would need intracellular concentrations that exist for 6.8 percent of the metabolites.

Two branches exist and the one described here is not the default. The abundance head is off unless an arm turns it on; with it off the layer falls back to a scalar availability $c_j \in [0,1]$ folded from the same γ_g and a nominal constant $\bar{E} = 10^{-3}$, giving $|\nu_j| \le c_j \sum_u k_{cat,u} \bar{E}$, and the budget is computed from the enzyme demand *implied* by a flux, $e_g = \sum_{u \ni g} |\nu_{\rho(u)}| / k_{cat,u}$, rather than from a predicted abundance. The measured consequence is in the flux-layer note: median capacity is 19.1 mmol gDW^{-1}h^{-1} under the constant, against 0.18 and 0.11 under the two abundance arms, and `feas_capacity_active_frac` is 0.0 for all three at initialization. **The bound is not yet observed to bind**, under either branch.

A measured proteome would replace E_g^0 rather than the head, and one exists in the same project: Zelezniak's 726 proteins across 97 kinase deletions. That is the change that would turn the prior from an equal-molar placeholder into a quantity.

Without Equation (4) the only limits on ν_j are yeast-GEM's own bounds, and those carry no capacity at all: **4,129 of 4,131 reactions sit at ± 1000 or 0**, which encodes direction and on-off and nothing else. A gene deletion would then have no route to change the *magnitude* of any flux.

$\Delta_r G$ enters as a term in the loss, not as a constraint. The idea, before any implementation detail, is two lines. A reaction should run downhill,

$$\nu_j \, \Delta_r G_j \;\le\; 0, \tag{6}$$

and where it does not, we charge for it:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{data}} \;+\; \lambda_{\text{th}} \sum_j \big[\, \nu_j \, \Delta_r G_j \, \big]_+. \tag{7}$$

The bracket is zero whenever Equation (6) holds and positive when it does not: no solver and no binary variable, and the tabulated formation energies are what make $\Delta_r G_j$ a measured quantity rather than a free parameter.

This is a training-time penalty, and at inference nothing enforces it. The flux still comes from the box, so a trained model can emit a thermodynamically infeasible flux; what training buys is a model that mostly does not. The layer therefore reports `feas_thermo_violation_frac`, the fraction of reactions with $\nu_j \Delta_r G_j > 0$, so the question is answered by measurement at evaluation rather than by assertion.

The form actually implemented adds a tolerance and a normalization:

$$\mathcal{L} = \mathcal{L}_{\text{data}} \; + \; \lambda_{\text{th}} \underbrace{\frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{\left[\, \nu_j \, \Delta_r G_j + \epsilon \,\right]_+}{|\nu_j| \, |\Delta_r G_j| + \epsilon}}_{C_{\text{th}}}, \qquad \Delta_r G_j = \Delta_r G_j'^{\circ} + RT \sum_{i \neq \mathrm{H}^+} S_{ij} \ln c_i, \tag{8}$$

where $\mathcal{J}$ runs over reactions with a known $\Delta_r G$, excluding exchange, biomass and water transport, and ϵ is a tolerance the strict inequality of Equation (6) needs once it is made smooth. The tabulated formation energies supply $\Delta_r G_j'^{\circ}$, which is the one place this document's thermodynamic table reaches the model.

The normalization is load-bearing, not cosmetic. The raw hinge $[\nu_j \Delta_r G_j]_+$ carries units and, measured at initialization on a real batch, is about **72 against a data loss of about 1**. An unnormalized term of that size does not regularize the model, it replaces it. Dividing by $|\nu_j||\Delta_r G_j|$ caps each reaction's contribution at 1 and turns C_{th} into the fraction of the available driving force running uphill, which is also the quantity worth reporting.

Why it earns its place: because $\Delta_r G$ is a difference of potentials it sums to zero around any cycle, so no cycle can run downhill at every step, and Equation (8) therefore penalizes internal loops. Capacity does not do this. Requiring growth at 90 percent of optimum narrows 2,316 reactions, 56.1 percent of the network, and leaves **427 reactions still free to take any value in** ± 1000, all of them internal loops. Only the thermodynamic term removes those, and it does so with no binary variable, which is what a loopless flux balance program needs one for.

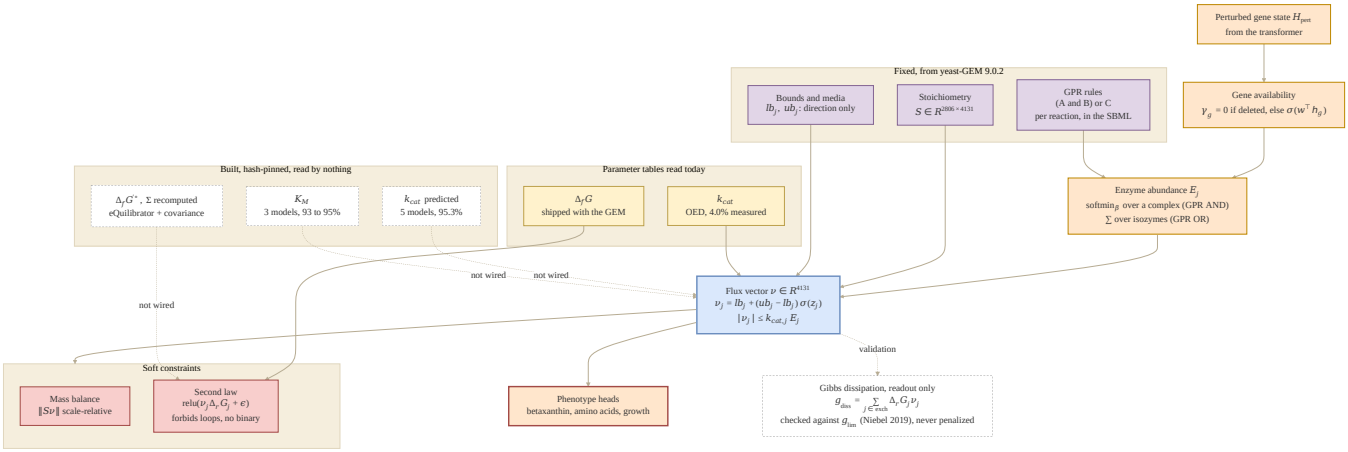


Figure 18: The flux module's main components and what feeds them. Orange is the path from the perturbed gene state to enzyme abundance; purple the fixed yeast-GEM 9.0.2 stoichiometry and bounds; yellow the parameter tables, with a **dashed** border and a dashed edge marking a table that is built and read by nothing; blue the flux vector; red the soft constraints. Every hard constraint is either exact by construction, as the box and the capacity bound are, or relaxed to a smooth penalty; nothing is enforced by a binary variable, which is what makes the layer differentiable and what removes the mixed-integer cost a per-genotype thermodynamic program pays. **Three parameter tables are drawn dashed because the drawing predates the wiring, and two of the three are now wired.** `resolve_kcat_table` takes a predicted tier between the measured values and the organism default, which fills 3,410 of the 3,728 catalytic units and leaves the 148 measurements untouched, and `resolve_reaction_delta_g` builds the standard reaction energies from a named route, the union of the shipped and recomputed metabolite tables pricing 3,722 reactions. K_M is still read by nothing. Source: `notes/torchcell.models.equivariant_cell_graph_transformer. mermaid.metabolic-module-simple.md`, rendered by `notes/assets/publish/scripts/mermaid_pdf.sh`. The full diagram, with the transformer stack and the Type I and Type II instruments, is not reproduced here because at this page width its labels print near 2 pt, under the 5 pt floor.

7 What these tables cannot support ✗

7.1 A predicted turnover number is not a measured one ✗

Every row is a model output, recorded as one in a provenance column. Two predictors trained on overlapping data, applied to the same 7,351 pairs, disagree by a median factor of about three, which is wider than either model's own reported error and is the honest uncertainty on any single value.

The accuracy of Section 4.6 does not narrow it, being an upper bound under training leakage. The join also has to be on the *pair*: joining on enzyme alone compares a prediction for one substrate against a measurement for another, and doing so moves one of these correlations from 0.98 to 0.82 and another from 0.62 to 0.12. The looser join is the one that is easy to write by accident.

7.2 K_M is built and deliberately not consumed ✗

K_M would enter the flux layer only through a saturation factor $\eta_{\text{sat}} = \prod_i c_i/(K_{M,i} + c_i)$, whose intracellular concentrations are measured for 191 of 2,806 metabolites, 6.8 percent, so adding it today adds a parameter and a free variable per reaction with no anchor for either.

The measured argument for adding it anyway is promiscuity: underground reactions show roughly twice the K_M at indistinguishable k_{cat}, so a k_{cat}-only model cannot tell a promiscuous side activity from a native one. The measured argument against is that no K_M predictor beats the EC-median lookup (Section 4.6), so these values are order-of-magnitude priors, not constants.

7.3 The thermodynamic gaps ✗

Four, in order of how much they bind.

Novel compounds needed a licensed tool, and no longer do. Uni-pKa on the simple enumeration template supplies the constants to 0.91 kJ mol^{-1} through `pkas_provided_externally` (Section 5.4). **What binds is reach rather than accuracy**: 347 of the 665 unpriced metabolites now carry a written structure and a price (Section 5.5), and 185 of the remaining 318 never can. The 48 values that came through the external route were written as neutral structures and priced with MolGpKa constants, so they still carry the 777.5 kJ mol^{-1} protonation-convention offset until they are resubmitted as their pH 7 major microspecies, which closes it to 0.13 kJ mol^{-1}.

Coverage is 66.4 percent against the shipped table's 84.9, up from 42.5 before the proton convention of Section 3, and 86.3 with the written lipid structures, which are at a single pH and not in the built table yet.

Proton translocation still carries zero in every arm reported here. `transport_delta_g_source: computed` is off by default, so the 183 proton-translocating reactions get none of the term Section 3.9 computes and checks, and the protonation contribution to N_H, worth 613 reactions, is still unwritten.

Compartment pH is assumed, and nothing in the library can settle it. The values are textbook organellar figures recorded as `assumed` in the build output; the two in-library sources tabulate rather than measure them and disagree by 1.2 units on the peroxisome (Section 3.7).

7.4 What the mirror records when a predictor cannot run ✗

All six now run. Where a predictor could not be run at all, the mirror records `runnable: false` with the specific blocker, so that a gap in coverage is never mistaken for a modeling choice. Boost_KM carried that flag wrongly (Section 1); the real obstacle was that UniRep is a TensorFlow 1 mLSTM stepping one residue at a time on CPU, which on a first attempt occupied roughly 85 cores, and the ESM-1b variant Wu et al. used embeds all 887 enzymes on one GPU in four minutes.

8 Every source, for citation and for the mirror ✗

Each predictor is a published method, a code repository at a pinned commit, and a weights archive, and the three are cited differently. Every commit and `sha256` comes from `$DATA_ROOT/models/kinetics/<name>/manifest.json` or from `mirror_kinetic_models.py`; the DOIs were verified against Crossref rather than recalled.

8.1 Papers, by the parameter they supply ✗

One table per parameter, because the question these are looked up to answer is always "where did *this* number come from", and that question is asked per parameter. Each row carries the mirrored artifact, its publication, how it was retrieved and the `sha256` the manifest pins it by, so a manifest field can be filled without re-deriving the provenance. `sha256` is given to the first eight characters of the one file that carries the prediction. Where the column reads "no manifest", the artifact sits outside `models/kinetics/` and has none, which is one of the defects Section 8.3 records.

Table 25: The six k_{cat} predictors: what was mirrored, the paper behind it, and the pin. Role is what the artifact does in this work.

Artifact	Publication	Retrieval	sha256	Role
DLKcat	Li et al., *Nat Catal* 5:662–672, 2022, 10.1038/s41929-022-00798-z	git clone SysBioChalmers/ DLKcat at 7c15d0d4	df5f6ab1	run over 3,552 units; base 2
UniKP	Yu et al., *Nat Commun* 14:8211, 2023, 10.1038/s41467-023-44113-1	git clone Luo-SynBioLab/ UniKP at 3ad5576a; regressors from HuggingFace HanselYu/UniKP	fe908112	run; second on rank accuracy
EITLEM	Shen et al., *Chem Catal* 4:101094, 2024, 10.1016/j.checat.2024.101094	git clone XvesS/ EITLEM-Kinetics at 7bb9d666; weights from Zenodo 10.5281/ zenodo.16153803, 12.3 GB	14 files pinned	run; best magnitude accuracy
TurNuP [6]	Kroll et al., *Nat Commun* 14:4139, 2023	git clone AlexanderKroll/ kcat_prediction at 98d88a37; Zenodo 8367052 and 8038678	da6cb47b	run; the one coverage outlier at 55.2%
DeepEnzyme	Wang et al., *Brief Bioinform* 25(5):bbae409, 2024, 10.1093/bib/bbae409	git clone hongzhonglu/ DeepEnzyme at d225ec31; figshare 10.6084/m9. figshare.25771062.v2	95962268	run; last on every accuracy measure
EZKinetics	unpublished, under review; no DOI	git clone le-yuan/ EZKinetics at bf7a8973, MIT since 2026-04-16; weights in repository	35de57b7	run; leads rank accuracy; the only condition-aware one
ESM-1b	Meta fair-esm; **publication not recorded in the mirror**	published fair-esm URL, after a silent bsdtar truncation	0569754e	enzyme encoder for TurNuP and for the Boost_KM variant Wu et al. [4] ran

Table 26: The four K_M predictors. Three of them are the same mirrors as Table 25, listed again with the file that carries the K_M model, because the artifact a K_M value traces to is not the one a k_{cat} value traces to.

Artifact	Publication	Retrieval	sha256	Role
Boost_KM	Kroll et al., *PLoS Biol* 19:e3001402, 2021, 10.1371/journal.pbio. 3001402	git clone AlexanderKroll/ KM_prediction at 432a4fc0	79833811	run; the paper's UniRep model, not the ESM-1b one Wu et al. [4] ran
UniKP	Yu et al., *Nat Commun* 14:8211, 2023, 10.1038/s41467-023-44113-1	as Table 25	f0b8d597	run; below the EC-median lookup on class plausibility
EITLEM	Shen et al., *Chem Catal* 4:101094, 2024, 10.1016/j.checat.2024.101094	as Table 25	14 files pinned	run; one of the two selected K_M predictors
EZKinetics	unpublished, under review; no DOI	as Table 25, ESM2_3B encoder rather than ESM2_650M	f23bd2fd	run; leads both K_M rankings

8.2 Code and weights ✗

8.3 Five provenance defects to repair ✗

These are recorded rather than fixed silently, because each one breaks the rebuild guarantee in a different way.

- **EITLEM's recorded source URL is dead.** The manifest holds XvQiao/EITLEM-Kinetics, which now returns 404; the repository was renamed to XvesS/EITLEM-Kinetics. The pinned commit still resolves there, so the mirrored bytes are fine and only the retrieval command needs updating.
- **DeepEnzyme's retrieval command is a placeholder.** It contains the literal text <figshare 10.6084/m9.figshare.25771062.v2> rather than a URL, so that one file cannot be re-fetched by running the record.
- **eQuilibrator and Yeast-MetaTwin have no manifest.** Both sit outside models/kinetics/. The MetaTwin commit had to be read out of .git, and metatwin.rar is pinned by size alone with no sha256.
- **All six kinetics manifests carry citation_key: null.** Once these papers are in the mirror, that field joins each weight file to its publication, which is what makes a predicted value citable rather than merely traceable.
- **The four pK_a predictors have no manifest, and three of their four publications are mirrored but not yet citable.** They were cloned and run, and Table 28 gives their license, repository and mirror key. A pK_a that reaches a formation energy is a model output like any other and needs the same pin, so these belong under models/ before the route is used for anything but a benchmark.

Table 27: Formation and reaction Gibbs energies. The two shipped CSVs are the incumbent this work compares against; the eQuilibrator route is what replaces them. The three rows below them are what the shipped CSVs are made of, listed separately because a value in either column traces to one of the three and not to yeast-GEM. **Neither shipped file has a provenance manifest of its own**, which is why their pins are the hashes recorded by `gem_audit.py` rather than by a mirror; the three upstream artifacts below them are mirrored with `sha256` in `thermo_yeast9_provenance.json`.

Artifact	Publication	Retrieval	`sha256`	Role
eQuilibrator 3.0 [10]	Beber et al., *Nucleic Acids Res* 50:D603–D609, 2022	`pip`, plus the compound cache at `~/.cache/equilibrator/compounds.sqlite`	no manifest	the recomputation: 2,113 metabolites, 2,741 reactions, with covariance
Component contribution	Noor et al., *PLoS Comput Biol* 9:e1003098, 2013, `10.1371/journal.pcbi.1003098`	method, inside eQuilibrator	no manifest	the estimator; why the table transfers to any organism
Shipped metabolite table	yeast-GEM 9.0.2 [1], curated from yETFL [8] by metabolite *name*, ModelSEED fallback on KEGG id	`data/databases/model_metDeltaG.csv`, shipped with the GEM	`179a5aac`	the incumbent formation table; 2,389 metabolites, **kcal mol^{-1}**, no uncertainty
Shipped reaction table	same, with a dG-predictor fallback attached by *row position*	`data/databases/model_rxnDeltaG.csv`, shipped with the GEM	`7a66ef5d`	the incumbent reaction column; 3,210 reactions, **two unit systems**
yETFL	Oftadeh et al. [8]	Zenodo `10.5281/zenodo.4778047`, `input_models.zip` and `info_yeast.zip`	`ec82ce09`, `3b96940a`	what both shipped columns hold: 3,177 of 3,210 reaction values verbatim, in kcal mol^{-1}
matTFA, Mavrovouniotis group contribution	Mavrovouniotis 1990, as cited by yETFL's Methods and packaged in matTFA's `thermo_data.thermodb` with Alberty's transform; **publication not recorded in the mirror**	through the yETFL model above; **not mirrored separately**	no manifest	the estimator behind yETFL: 298.15 K, per-compartment pH and I, no magnesium
dG-predictor	Wang et al. 2021; **publication not recorded in the mirror**	`data/model Curation/v8_7_0/` of yeast-GEM, with `run_dgpredictor.py`	`0870ec4d`	the fallback on 33 reactions, in kJ mol^{-1}, attached by row position
Yeast-MetaTwin [4]	Wu et al., *Nat Catal* 9:434–447, 2026	`git clone LiLabTsinghua/Yeast-MetaTwin` at `de1268ee`; Zenodo `10.5281/zenodo.13911783`, 6.6 GB	no manifest	the figure being reproduced, and the ESM-1b Boost_KM variant

One extraction hazard is worth recording alongside them. `bsdtar` reading `esm1b_t33_650M_UR50S.pt` out of the 6.6 GB RAR **exited 0 and wrote 649,783,100 bytes** where the archive's own listing says 7,828,576,466. The truncation is silent, and a checkpoint that short still opens far enough to look like a file. It was caught by comparing against the listed size and then confirmed by `torch.load` refusing it. The bytes finally used were fetched from the published `fair-esm` URL and verified to the recorded `sha256`,

Table 28: Acid dissociation constants, which is the input that decides whether a compound outside the cache can be priced at all (Section 5.4). The first row is the incumbent and the reference the other four are scored against; it is the only one that is not open source. Three of the four publications reached the paper mirror on 2026-09-10, under the citation keys given, and none of them is in this document's served bibliography yet, so they are named rather than cited. **None of these has a mirror manifest yet**, since they sit outside `models/kinetics/`; the pin is the repository commit, or the weight revision where the weights are not in the repository.

Artifact	Publication	Retrieval	Pin	Role
ChemAxon `cxcalc`	proprietary; reached through eQuilibrator's compound cache	not installed here; the cache's stored constants are used	the cache file	the reference the other four are scored against; the license that blocks new compounds
`equilibrator-assets` master	Beber et al. [10]	`git clone gitlab.com/equilibrator/equilibrator-assets`	`c79e371e`	the bridge: `pkas_provided_externally`, absent from 0.6.0
Uni-pKa	Luo et al., *JACS Au* 4:3451–3465, 2024, mirrored as `luoBridgingMachineLearning2024`, with the preprint as `luoUnipKaAccuratePhysically`; Apache-2.0	`pip install unipkainfer`; five folds `fold_{0..4}/checkpoint_best.pt`, 571,250,635 bytes each, from HuggingFace `bindresearch/pkasso_unipka`; GPU	HuggingFace revision `e28a5917`; `fold_0` sha256 `4d5a5fc9...062a200`	**the tool selected**, on the simple enumeration template: 0.91 kJ mol^{-1}, 299 of 300
MolGpKa	Pan et al., *J Chem Inf Model* 61:3159–3165, 2021, mirrored as `panMolGpkaWebServer2021`; MIT, though the license header names OpenNMT as copyright holder rather than the authors	`git clone`; weights in repository, 6.0 MB each; CPU	repository commit	1.21 kJ mol^{-1}, ran all 300; flagged on 48.3 percent of ladders
pKasolver-lite	**publication not yet recorded**; MIT	`git clone`; 50 weight files, 105 MB; needs PyG exactly 2.0.1	repository commit	2.52 kJ mol^{-1}; degrades to 7.13 on six-constant ladders
QupKake	Abarbanel and Hutchison, *J Chem Theory Comput* 20:6946–6956, 2024, mirrored as `abarbanelQupKakeIntegratingMachine2024`; BSD-3	`git clone`; ships its own xTB 6.4.1 binary	repository commit	**unmeasured**: 4 of up to 14 rungs, 1 of 60 ladders finished

Table 29: What was mirrored, at which revision. A commit is given in full because a short hash is not a pin.

Tool	Repository, at commit	Weights
DLKcat	`SysBioChalmers/DLKcat` `7c15d0d4a7ac029f9d75564d9f2a93874aeaaec7`	in repo, extensionless `state_dict`
UniKP	`Luo-SynBioLab/UniKP` `3ad5576aaa2c8c0dd0e0b6c283c1d365ab23c6ea`	HuggingFace `HanselYu/UniKP`
EITLEM	`XvesS/EITLEM-Kinetics` `7bb9d6668d86263f17dddc1915fccff5712fd511`	Zenodo `10.5281/zenodo.16153803`, 12.3 GB
TurNuP	`AlexanderKroll/kcat_prediction` `98d88a37b66f64381086b5492f315e763c528fa0`	Zenodo `10.5281/zenodo.8367052` and `10.5281/zenodo.8038678`
DeepEnzyme	`hongzhonglu/DeepEnzyme` `d225ec31a1079177eea38fcf44c5c4b7afe516d8`	figshare `10.6084/m9.figshare.25771062.v2`
Boost_KM	`AlexanderKroll/KM_prediction` `432a4fc02afa20fcb8296fb96ce035a99ae33e85`	in repo; the ESM-1b variant is vendored in the Yeast-MetaTwin release
ESM-1b	Meta `fair-esm`, base checkpoint	`esm1b_t33_650M_UR50S.pt`, sha256 `0569754e...83a8b1`
Yeast-MetaTwin	`LiLabTsinghua/Yeast-MetaTwin` `de1268eea271a67ffb0581aab51b3549c8020950`	Zenodo `10.5281/zenodo.13911783`, 6.6 GB

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